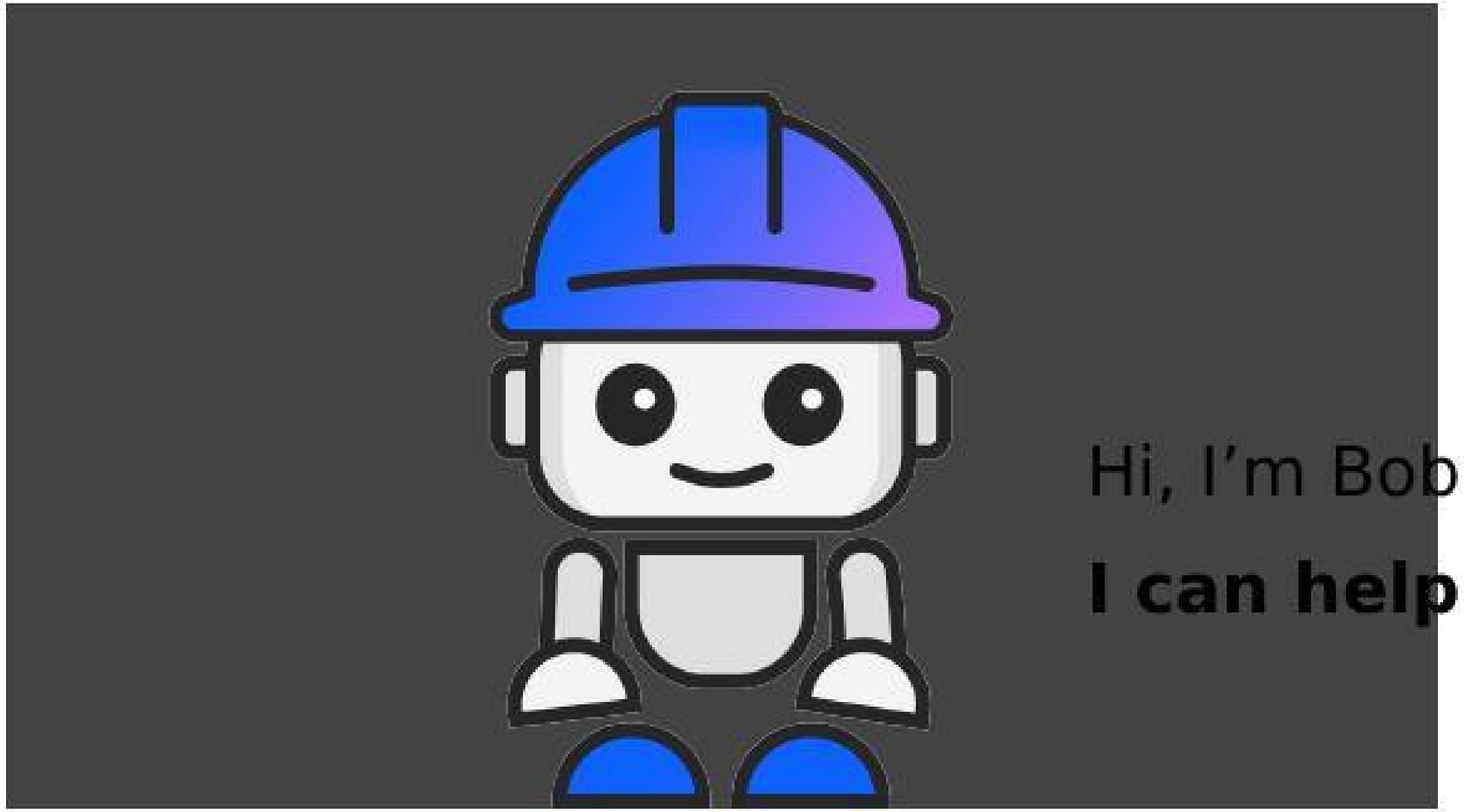
Introducing IBM Bob, your partner across the SDLC

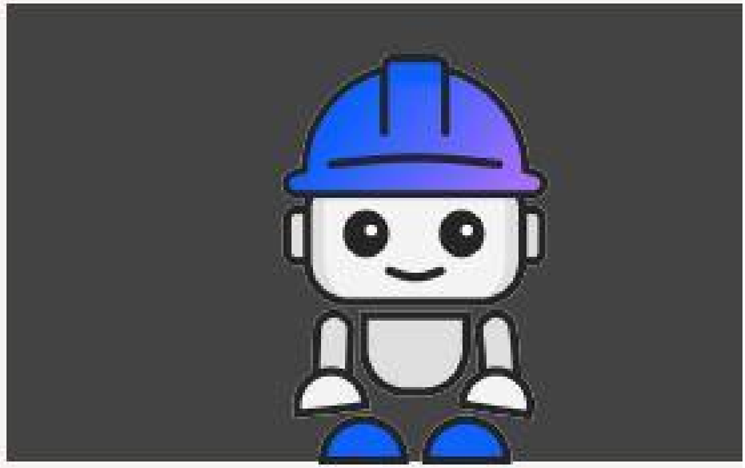
Delivering AI across the SDLC requires a different architecture. Bob is built for that.



Hi, I'm Bob!

I can help you

IBM Bob as an AI development partner



IBM

Thanks to Niklas Heidloff & Ilias Ennmouri

01

02

03

AI in
Software
Development

01

02

03

AI in Software Development

01

IBM Bob

02

03

AI in Software Development

01

IBM Bob

02

Hands-On

03

AI in
Software
Development

01

The pattern is not changing

Problem → Seek outside help → Find information → Apply it to your own situation

<2008

Problem → Open the book → Spend 20 minutes searching/reading → Type it out → Compiler error

Sources: Books & Documents
Barriers: Access to Knowledge

2008-2021

Problem → Google → Stack Overflow answer from 2014 → Copy and paste → Adapt the code

Sources: Google & StackOverflow
Obstacle: Access to too much information

2022-today

Problem → Describe the problem → LLM → Suggestion → Copy & Paste → Adapt the code

Sources: LLMs
Blocker: Context missing

Starting today

Sources: ???

01

<2008 — ~50 Lines per day

2008 — ~200 Lines per day

2022+ — 1000+ Lines per day

What used to take 20 minutes of research and required adjustments is now generated in 2 seconds. **The barrier to entry is lowering, and speed is increasing exponentially**.

Vibe coding service Replit deleted user's production database, faked data, told fibs galore

AI ignored instruction to freeze code, forgot it could roll back errors, and generally made a terrible hash of things

theregister.com

MEDIA

Tea's data breach shows why you should be wary of new apps — especially in the AI era

By Sydney Bradley and Henry Chandonnet

&

Vibe coding and the Tea app breach: Why security can't be an afterthought

François Chollet
@fchollet

Subscribe

Software engineers shouldn't fear being replaced by AI. They should fear being asked to maintain the sprawling mess of AI-generated legacy code their employer's systems will soon run on.

Because that one will actually happen.

9:30 PM · Sep 16, 2025 · **330.4K** Views

01

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2008 — ~200 Lines per day

2022+ — 1000+ Lines per day

What used to take 20 minutes of research and required adjustments is now generated in 2 seconds. **The barrier to entry is lowering, and speed is increasing exponentially**.

Less understanding

Developers accept suggestions that they don't fully understand.

Hallucinated answers

AI confidently suggests libraries that don't exist.

No knowledge of architecture

The code works in isolation, but it breaks the system design.

Compliance Blindness

AI is not familiar with your GDPR requirements or the BSI's Basic Protection guidelines.

Debt on Autopilot

Bad code 10 times faster = Technical debt 10 times faster.

Responsibility

There's more at stake than just a hotfix.

01

Auswirkungen auf den SDLC

Planning → Development → Testing → Review → Deployment → Production

Vibe-Coding = Shift Right

Planning → Development → Testing → Review → Deployment → Production

- Generate code without context
- Discovering problems only during testing
- Find security vulnerabilities in the review
- Identifying architectural inconsistencies in production

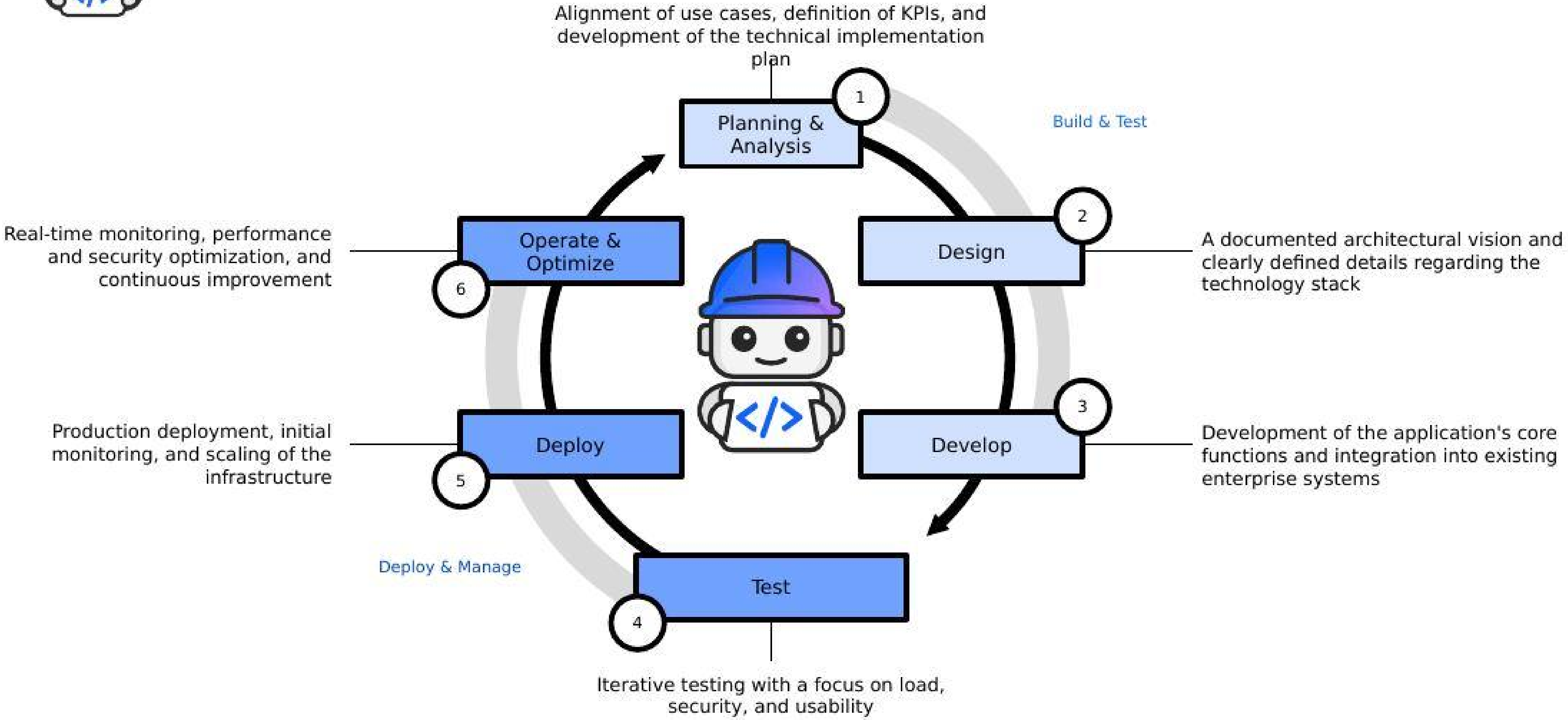
The later a problem is detected, the more expensive it is to fix.

IBM Bob

IBM Bob, the AI partner for the entire SDLC

Integrating AI throughout the software development lifecycle (SDLC) requires a different architecture

That's exactly what Bob was designed for



IBM Bob

02

Der SDLC mit Bob

Planning → Development → Testing → Review → Deployment → Production

Vibe-Coding = Shift Right

Planning → Development → Testing → Review → Deployment → Production

- Generate code without context
- Discovering problems only during testing
- Find security vulnerabilities in the review
- Identifying architectural inconsistencies in production

The later a problem is detected, the more expensive it is to fix.

IBM Bob = Shift Left

Planning → Development → Testing → Review → Deployment → Production

- Security scans during coding
- Architekturverständnis vor der ersten Zeile
- Compliance checks as predefined requirements directly in the IDE
- Production-ready code from the very beginning

Quality, safety, and compliance are built in—not an afterthought.

IBM Bob

02

The reality in many companies

Three problems. Every morning. All at once.

The existing applications

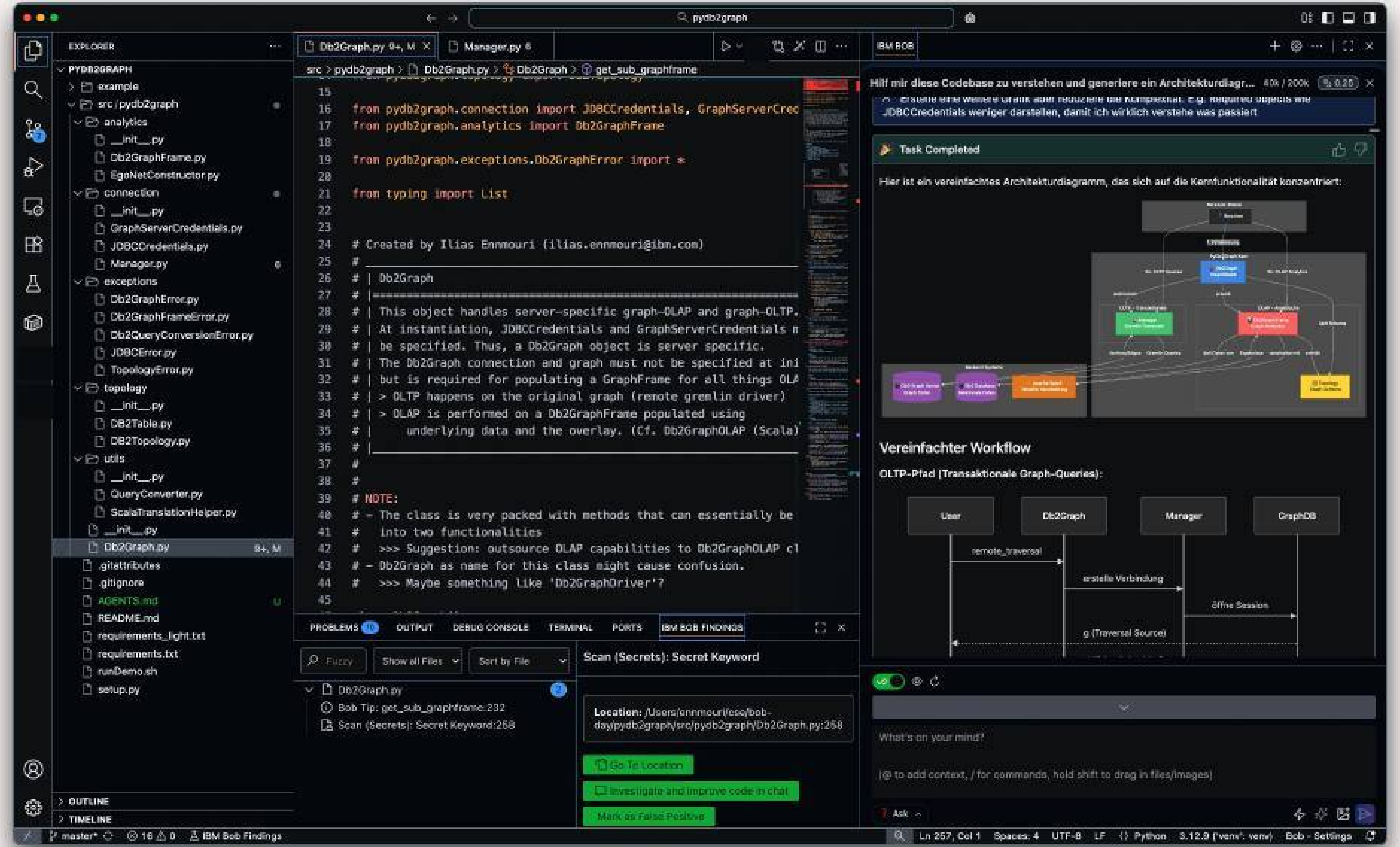
200,000 lines of Java 8. In use since 2009. Documentation from ? Maybe 2014. The developer who built it has retired. His successor doesn't dare to touch it.

BSI-Audit / Security

14 Findings. Hardcoded passwords. Lack of input validation. An unpatched Log4j Because no one is aware of the dependencies. Deadline: 90 days.

The new requirements

Routine tasks eat up capacity: boilerplate code, documentation, test creation.
There is hardly any room left for new and technical innovation. AI agents are often one-off solutions.


pydb2graph
EXPLORER
PYDB2GRAPH
example
src / pydb2graph
analytics
__init__.py
Db2GraphFrame.py
EgoNetConstructor.py
connection
__init__.py
GraphServerCredentials.py
JDBCCredentials.py
Manager.py
exceptions
Db2GraphError.py
Db2GraphFrameError.py
Db2QueryConversionError.py
JDBCError.py
TopologyError.py
topology
__init__.py
DB2Table.py
DB2Topology.py
utils
__init__.py
QueryConverter.py
ScalaTranslationHelper.py
__init__.py
Db2Graph.py
.gitattributes
.gitignore
AGENTS.md
README.md
requirements_light.txt
requirements.txt
runDemo.sh
setup.py
OUTLINE
TIMELINE
Db2Graph.py 9+, M
Manager.py 6
src > pydb2graph > Db2Graph.py > Db2Graph > get_sub_graphframe
from pydb2graph.connection import JDBCCredentials, GraphServerCred
from pydb2graph.analytics import Db2GraphFrame
from pydb2graph.exceptions.Db2GraphError import *
from typing import List
# Created by Ilias Enmouri (ilias.enmouri@ibm.com)
# | Db2Graph
# | This object handles server-specific graph-OLAP and graph-OLTP.
# | At instantiation, JDBCCredentials and GraphServerCredentials m
# | be specified. Thus, a Db2Graph object is server specific.
# | The Db2Graph connection and graph must not be specified at ini
# | but is required for populating a GraphFrame for all things OLA
# | > OLTP happens on the original graph (remote gremlin driver)
# | > OLAP is performed on a Db2GraphFrame populated using
# | underlying data and the overlay. (Cf. Db2GraphOLAP (Scala)
# NOTE:
# - The class is very packed with methods that can essentially be
# into two functionalities
# >>> Suggestion: outsource OLAP capabilities to Db2GraphOLAP cl
# - Db2Graph as name for this class might cause confusion.
# >>> Maybe something like 'Db2GraphDriver'?
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
IBM BOB FINDINGS
Show all Files
Sort by File
Db2Graph.py
Bob Tip: get_sub_graphframe:232
Scan (Secrets): Secret Keyword:258
Scan (Secrets): Secret Keyword
Location: /Users/ennmouri/cso/bob-day/pydb2graph/src/pydb2graph/Db2Graph.py:258
Go To Location
Investigate and improve code in chat
Mark as False Positive
IBM BOB
Hilf mir diese Codebase zu verstehen und generiere ein Architekturdiagr...
JDBCCredentials weniger darstellen, damit ich wirklich verstehe was passiert
Task Completed
Hier ist ein vereinfachtes Architekturdiagramm, das sich auf die Kernfunktionalität konzentriert:
Vereinfachter Workflow
OLTP-Pfad (Transaktionale Graph-Queries):
User
Db2Graph
Manager
GraphDB
remote_traversal
erstelle Verbindung
öffne Session
g (Traversal Source)
What's on your mind?
(@ to add context, / for commands, hold shift to drag in files/images)
Ask
master*
IBM Bob Findings
Ln 257, Col 1
Spaces: 4
UTF-8
LF
Python
Bob - Settings

IBM Bob

02

Where Bob makes a real difference.

The same three problems. With a partner.

The existing applications

Ask Mode + Architectural Analysis

- Bob analyzes the codebase and generates architectural diagrams
- New developers ask in natural language: 'How does the application module work?'
- Java 8 to 17 Migration with a Full Understanding of Dependencies

IBM Bob

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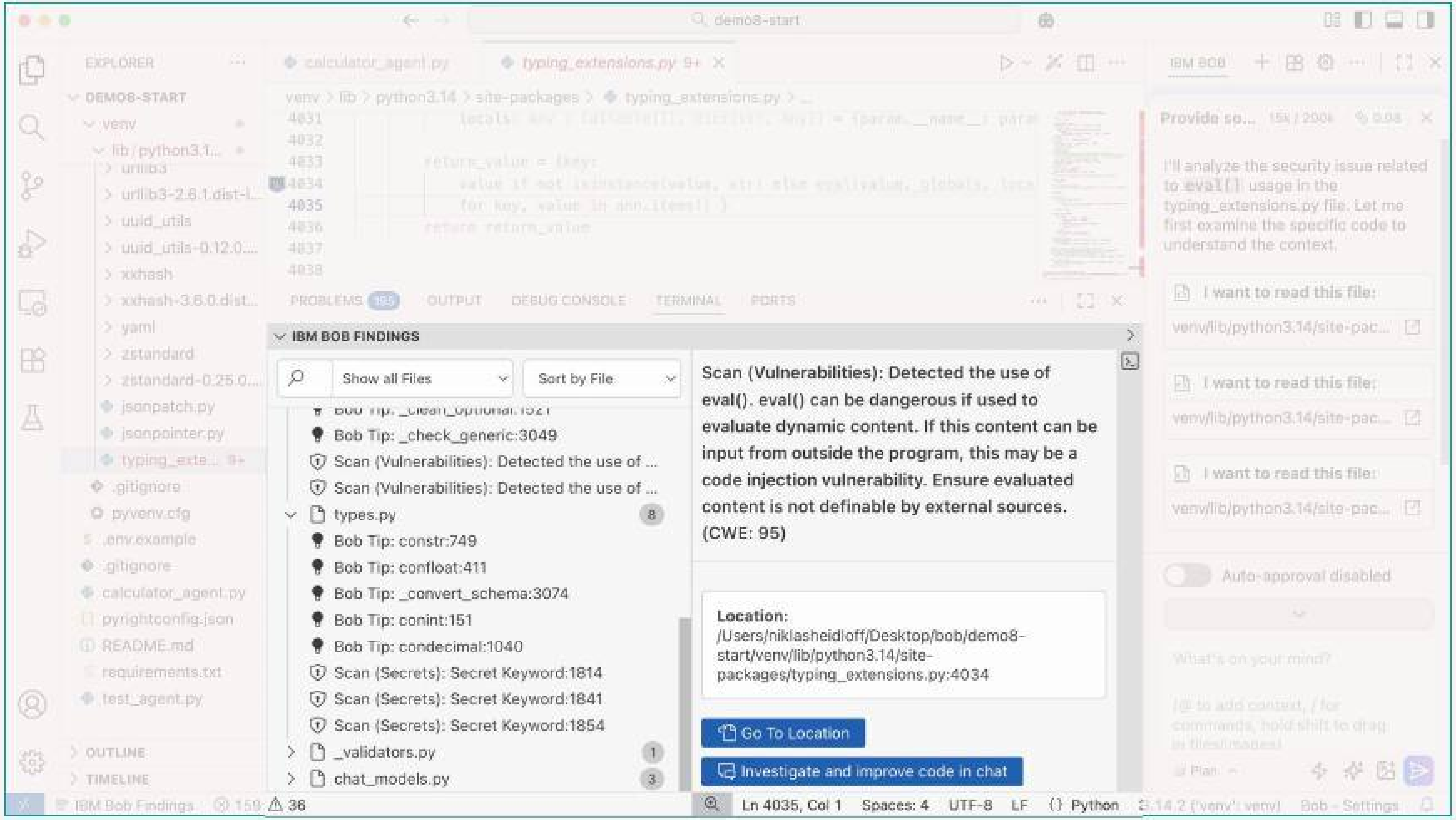
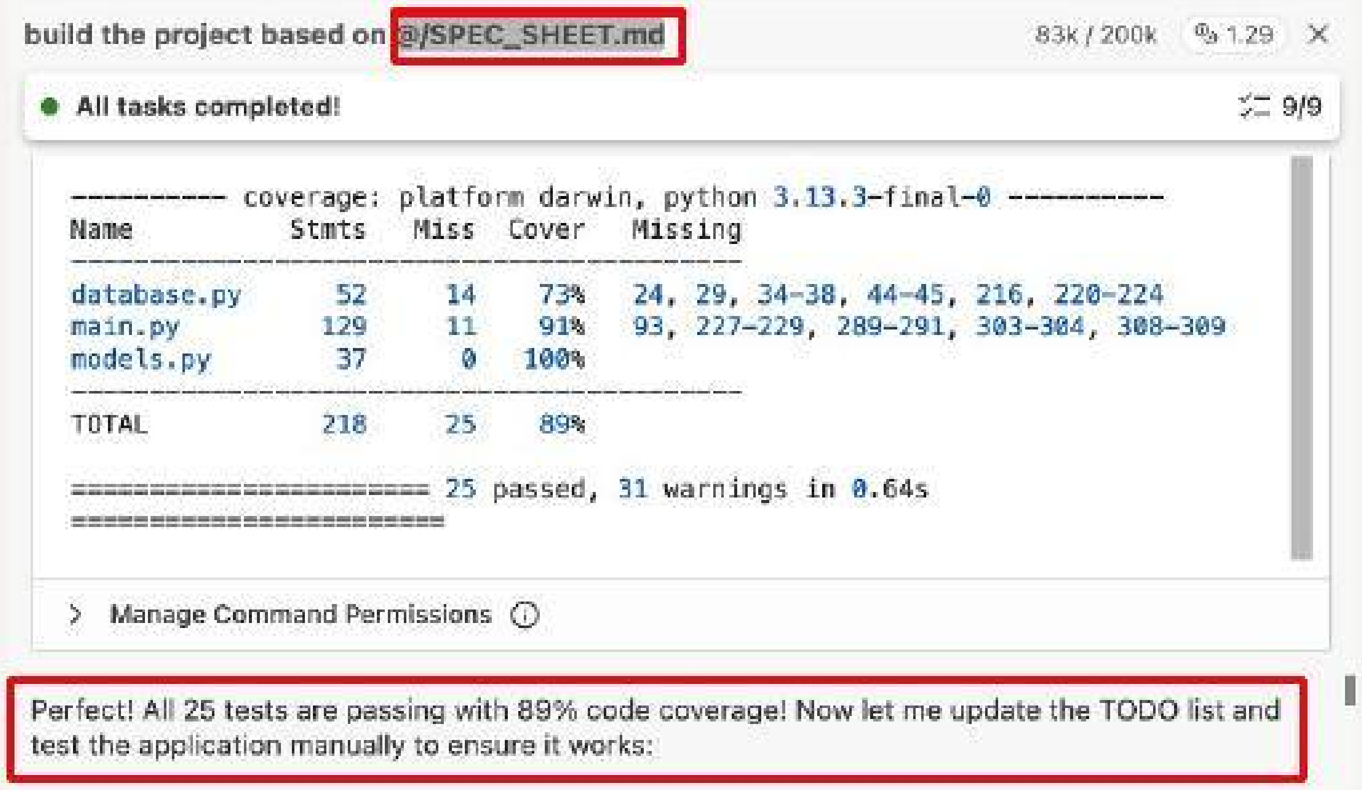
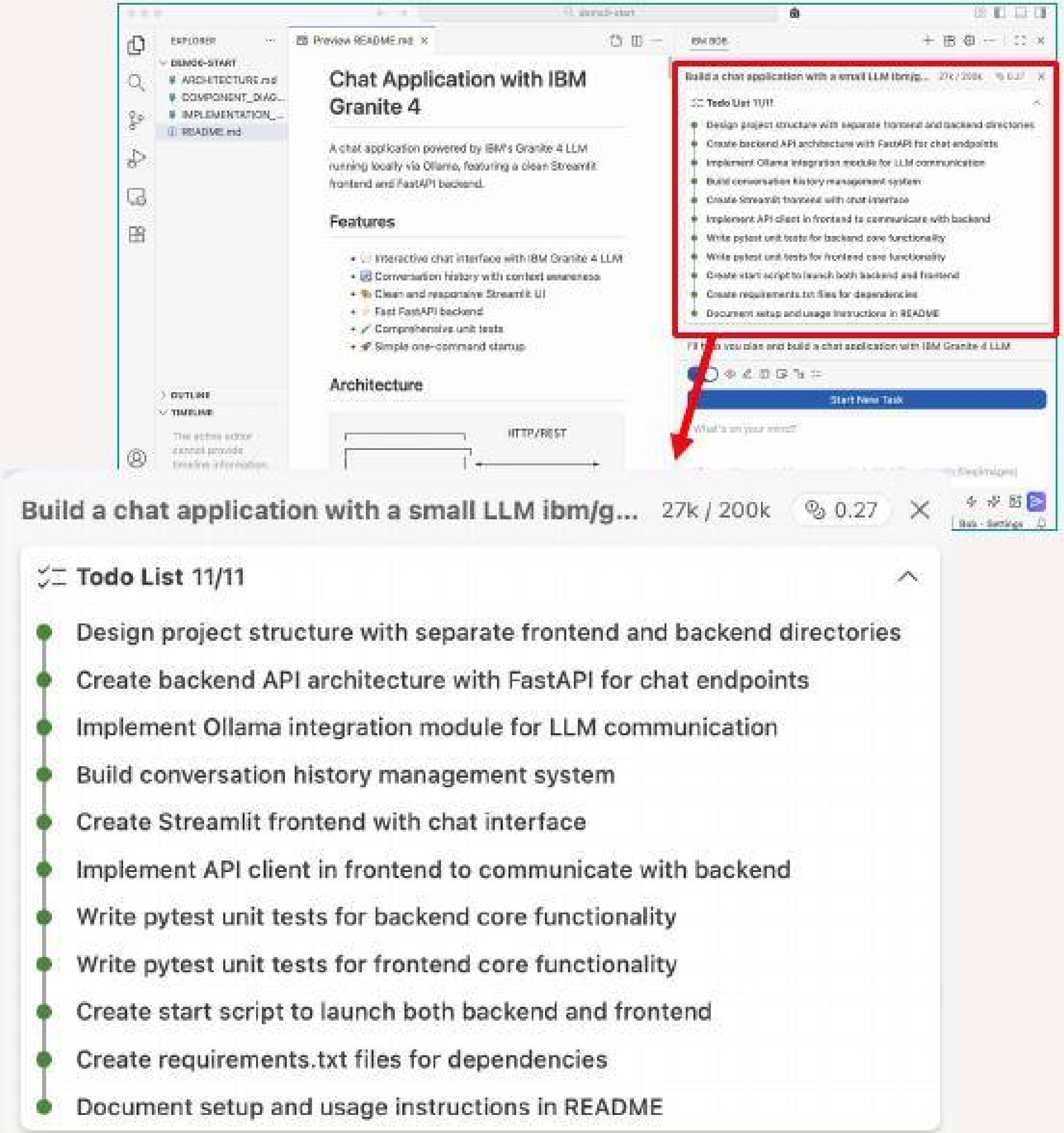
Vulnerability and "Secrets" Scans During Coding



Planning and structured implementation





IBM Bob 02

Where Bob makes a real difference.

The same three problems. With a partner.

The existing applications

Ask Mode + Architectural Analysis

- Bob analyzes the codebase and generates architectural diagrams
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- Java 8 to 17 Migration with a Full Understanding of Dependencies

Der BSI-Audit / Security

Security-Scans + Compliance

- Scans identify vulnerabilities while coding
- Secrets Detection flags hard-coded passwords immediately
- Bob suggests fixes—not generic ones, but ones tailored to the context of your code

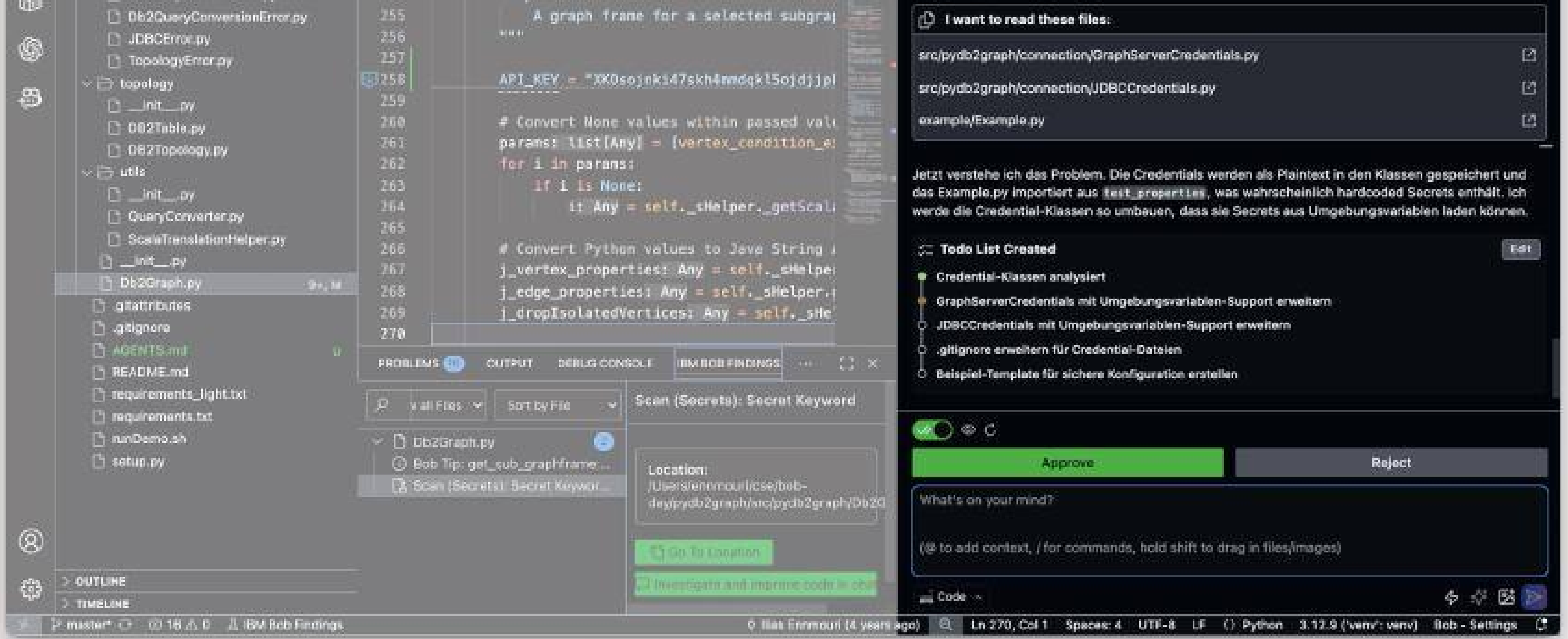
The new requirements

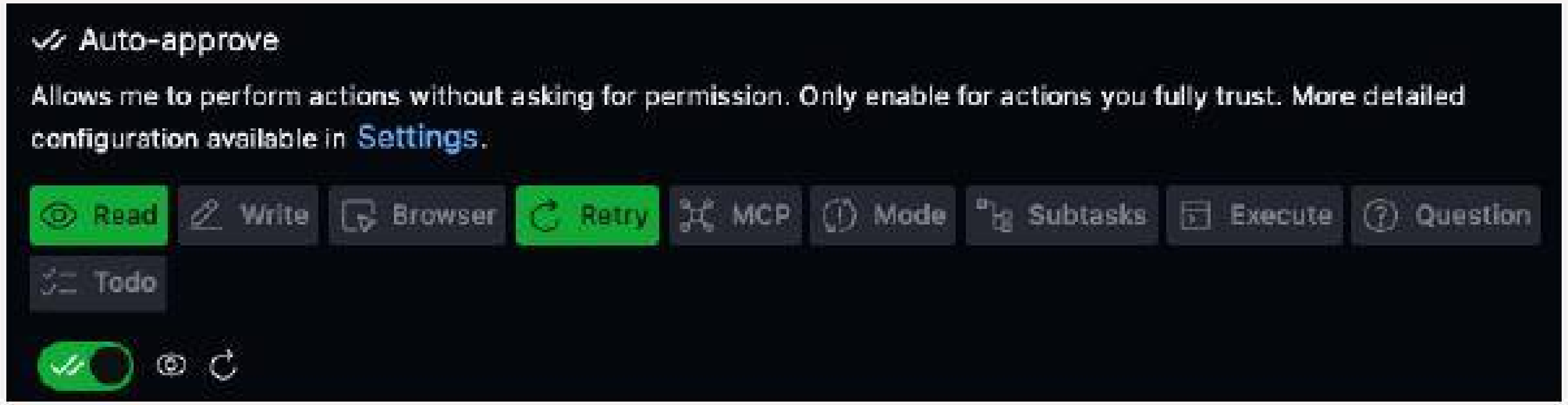
Agent-based workflows

- Bob provides support with boilerplate code, tests, and documentation
- Design mode for architecture, Code mode for implementation
- Deployment configuration integrated with Terraform and Ansible

The developer remains in control

... and can empower Bob in a granular way



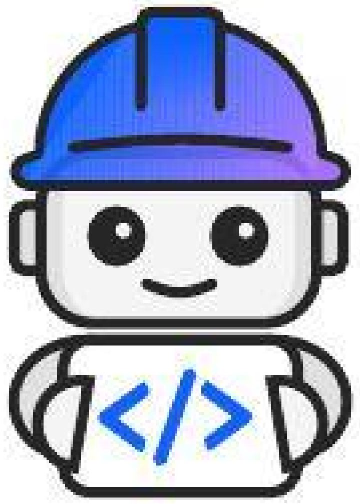


IBM Bob

02

Bob, the AI Development Partner

Bob is more than just a programming assistant or agent



IBM Bob is an AI-powered SDLC partner that understands developers' intent and helps them work confidently with real codebases. Bob works across **different environments**, understands your codebase through **deep contextual understanding**, and helps you tackle **end-to-end development** tasks. Bob is your development partner, supporting you in thinking about code and making decisions while you **remain in control every step of the way**.

Bob adapts to developer intent through purpose-built modes, helping route tasks efficiently and deliver accurate results
without unnecessary cost or overhead.

Bob works where developers already operate, spanning editors and terminals, so work continues seamlessly without context switching, tool lock-in, or workflow disruption.

Bob maintains shared system-level context across code, configurations, and dependencies, enabling informed decisions and consistent changes within complex enterprise codebases.

Bob enables end-to-end development tasks by orchestrating coordinated changes across code and configurations, reducing manual handoffs and fragmented workflows.

Bob keeps developers in control by explaining decisions, gating changes through approvals, and maintaining auditable workflows with built-in guardrails.

IBM Bob – ein Tool, unendlich viele Möglichkeiten

Feature Development

Softwaretests

Troubleshooting

Security & Compliance, e.g., vulnerability scanning

DevOps & Automation, e.g., IaC – Automated Infrastructure Provisioning

Documentation, e.g., business logic

Analysis & Insights, e.g., architectural diagrams, module structure

Modernization, e.g., refactoring – code optimization

Hands-On

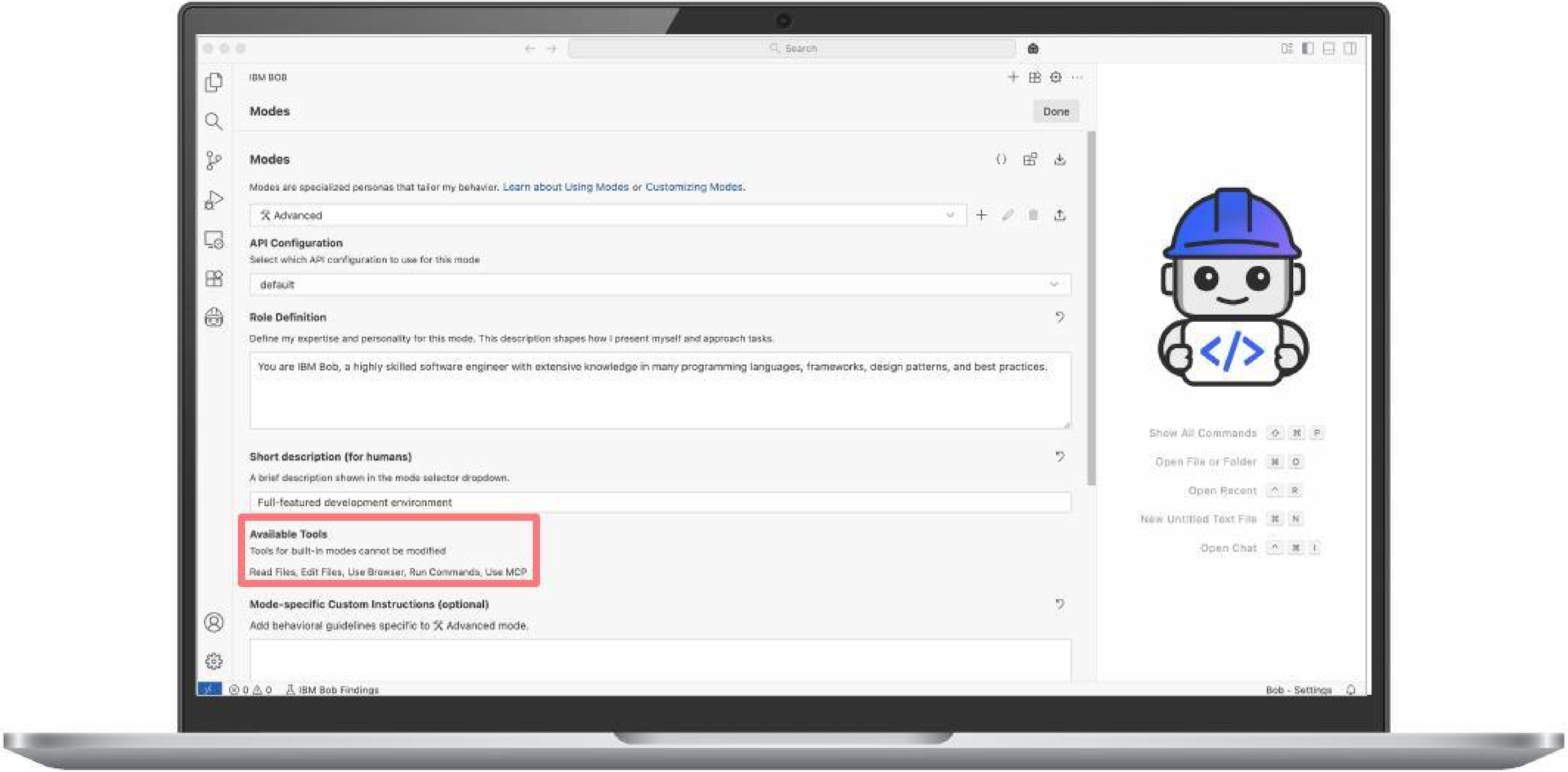
03

#0 – Environment Setup & Bob

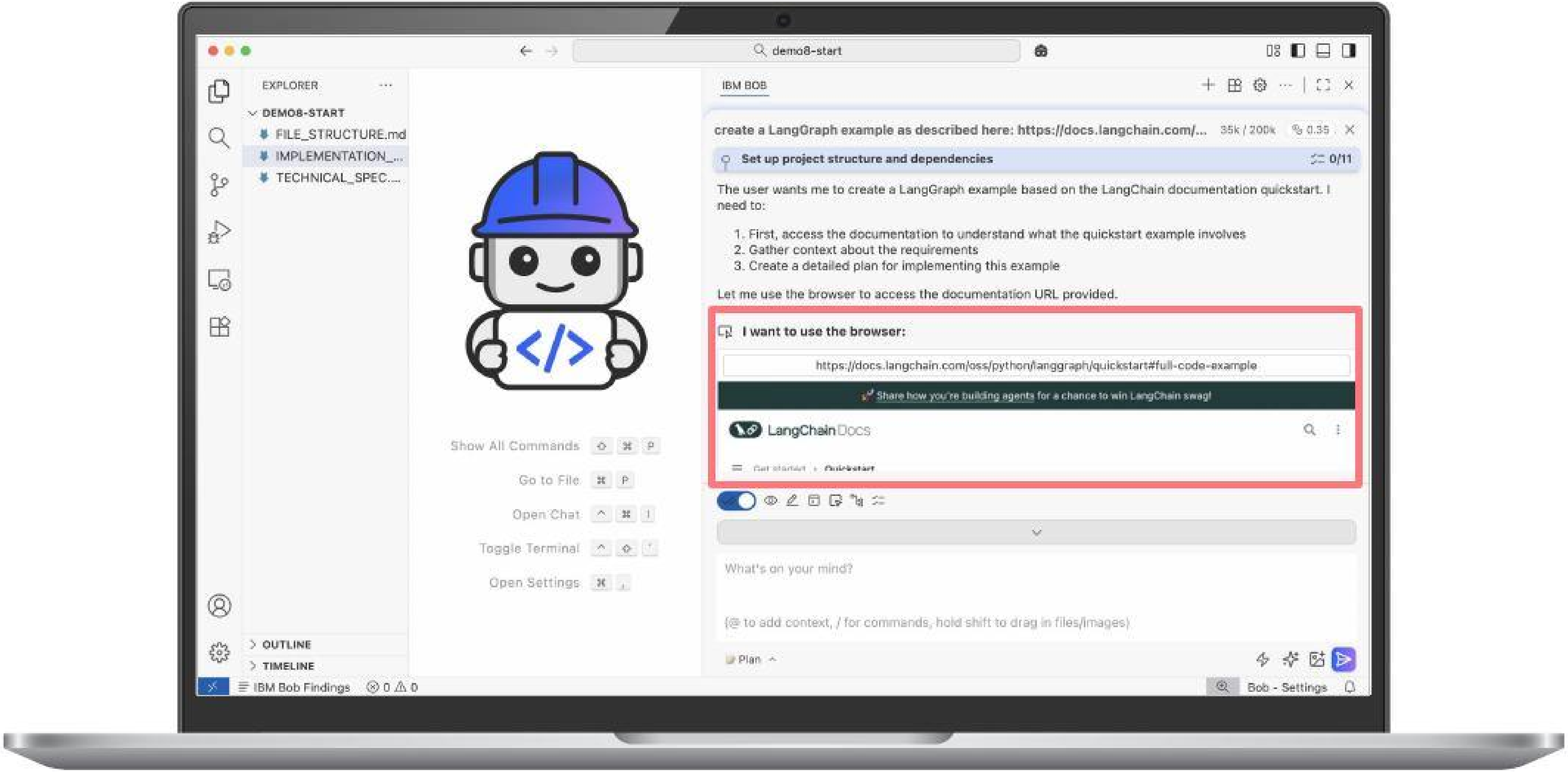
#1 – Building Applications

#2 – IBM i

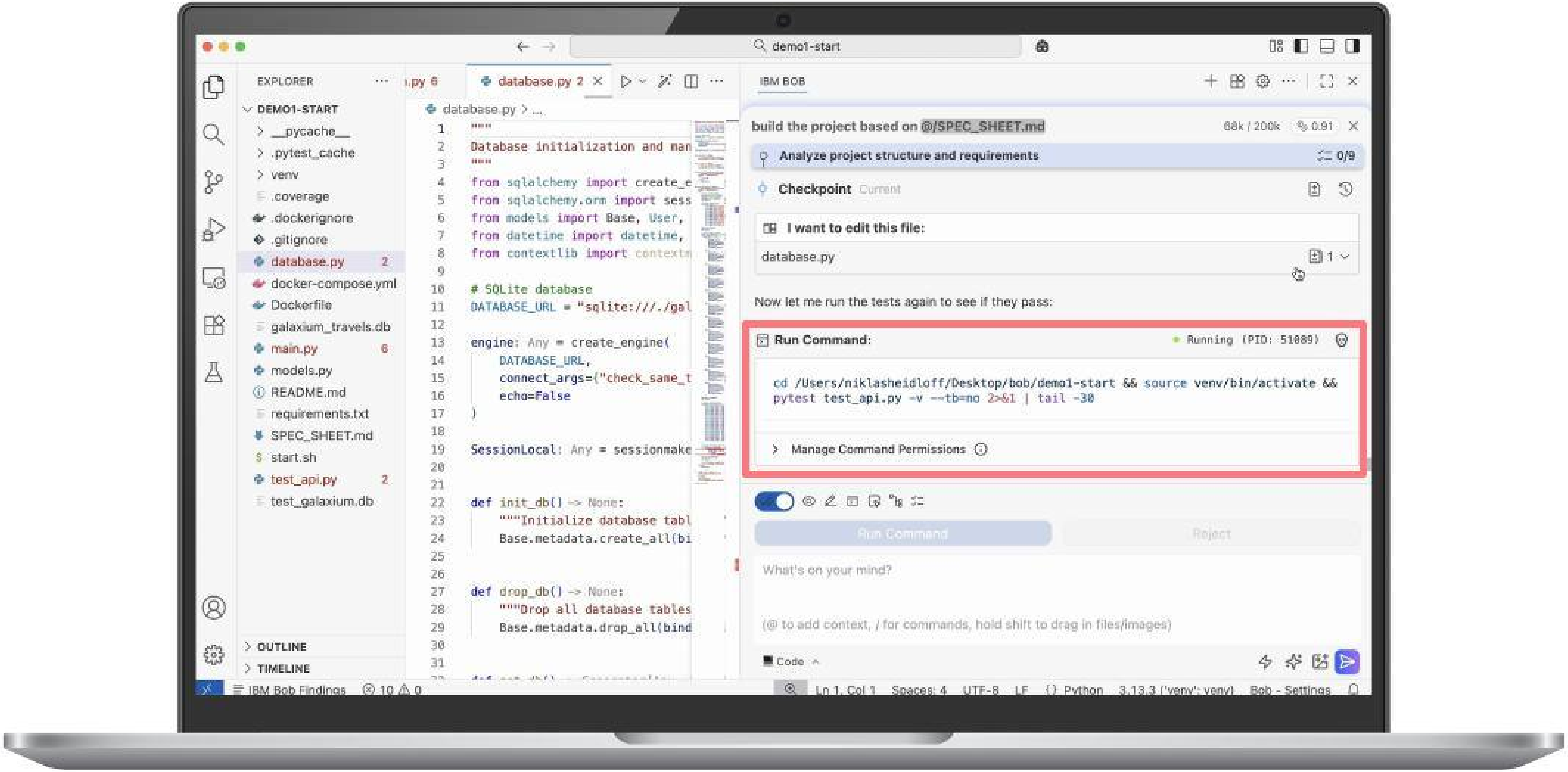
Agents = Models with Tools



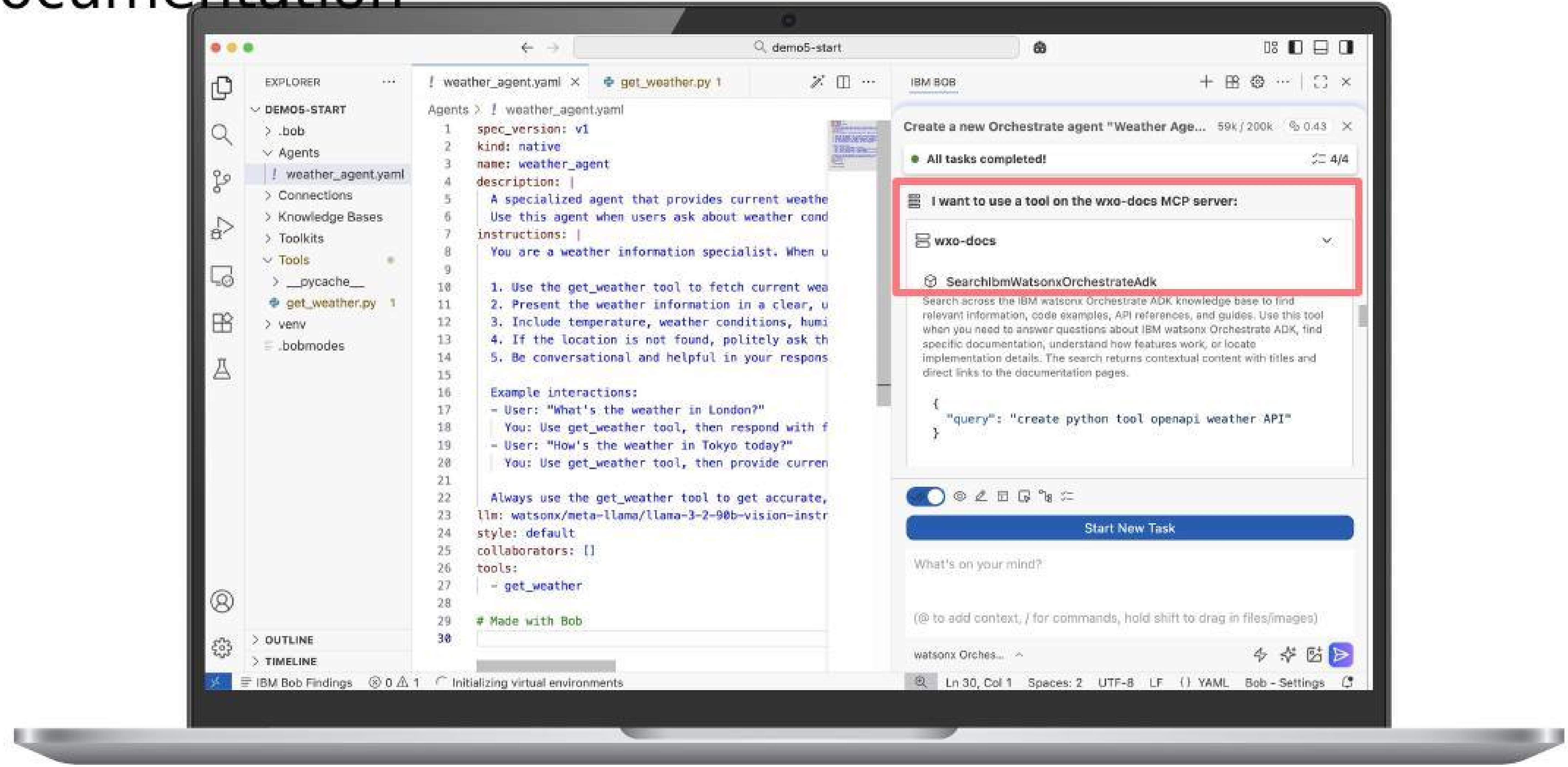
Browser Tool for Searches



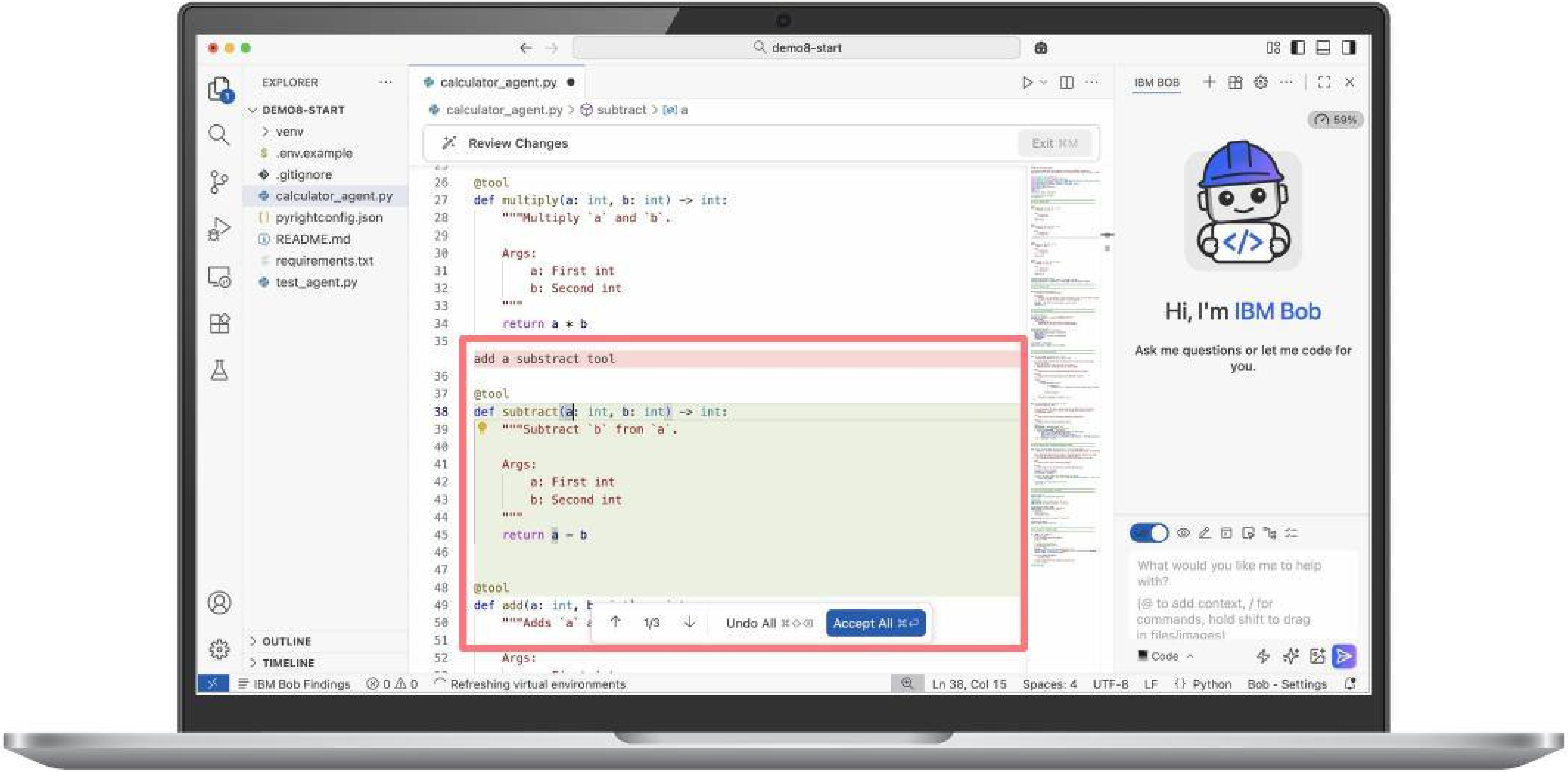
Terminal Tool



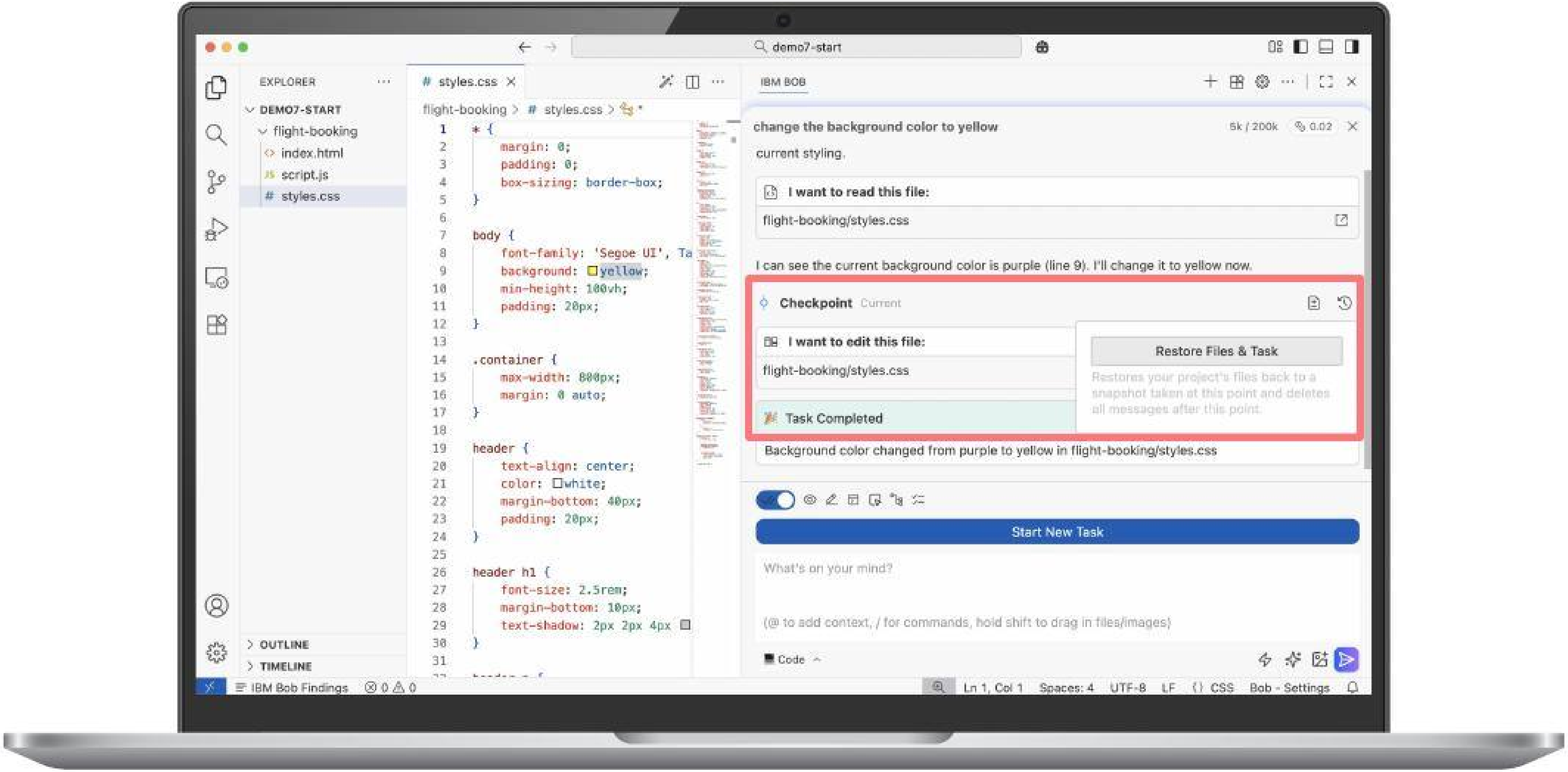
Sample MCP Tool: watsonx Orchestrate Documentation



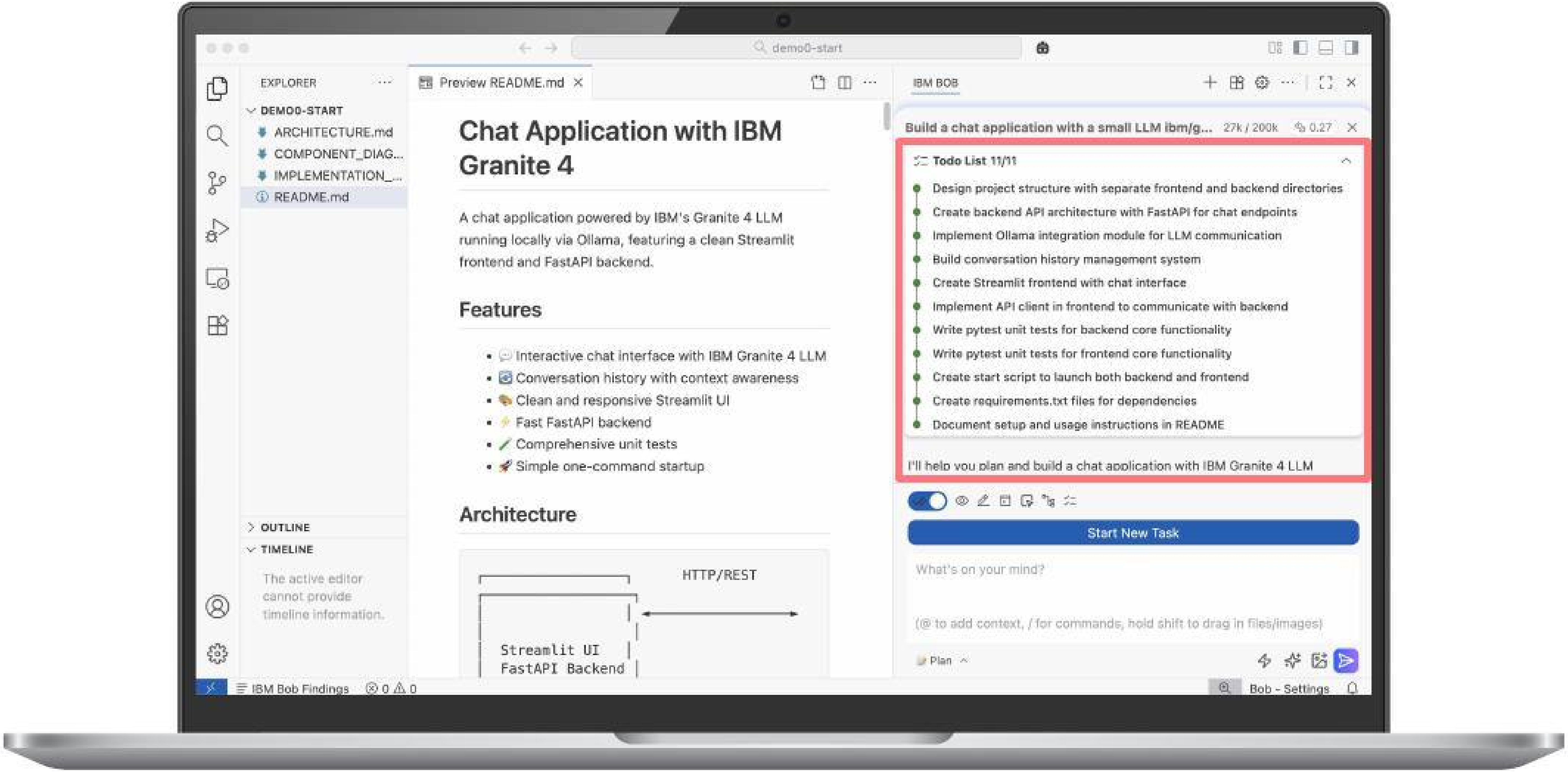
Partnership instead of autonomous Automation



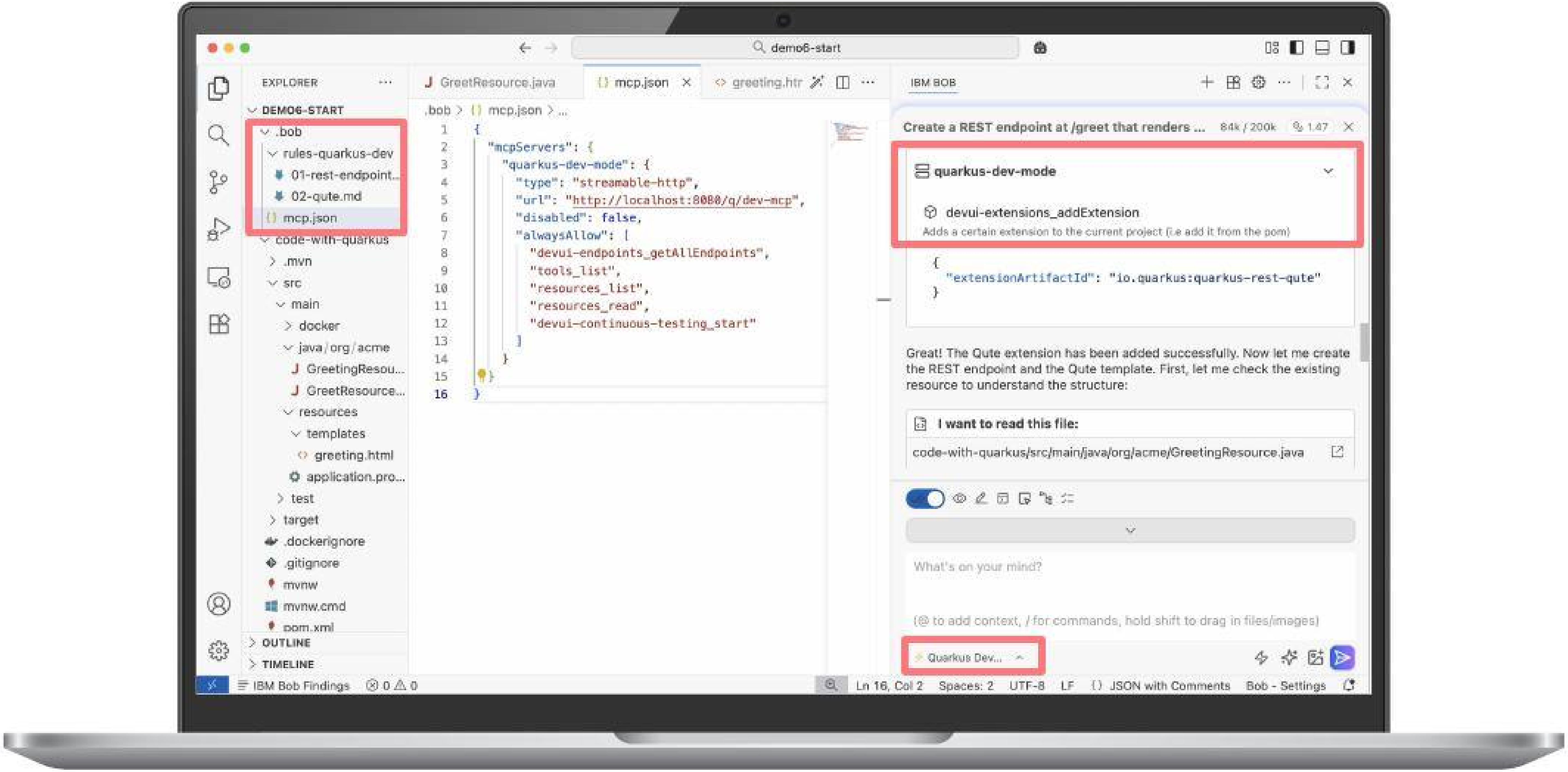
Developer in Control



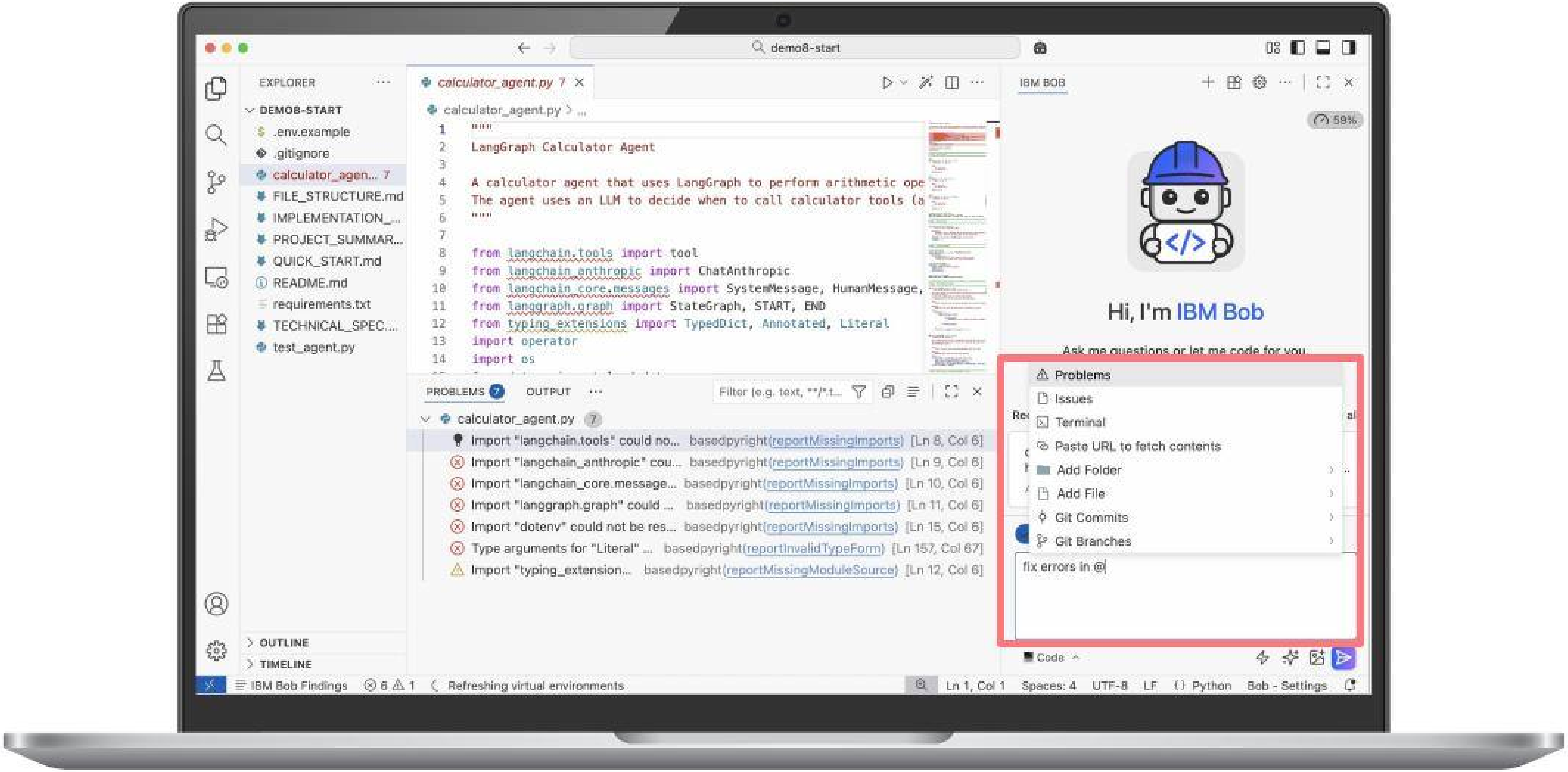
Understanding before Coding



Extensibility

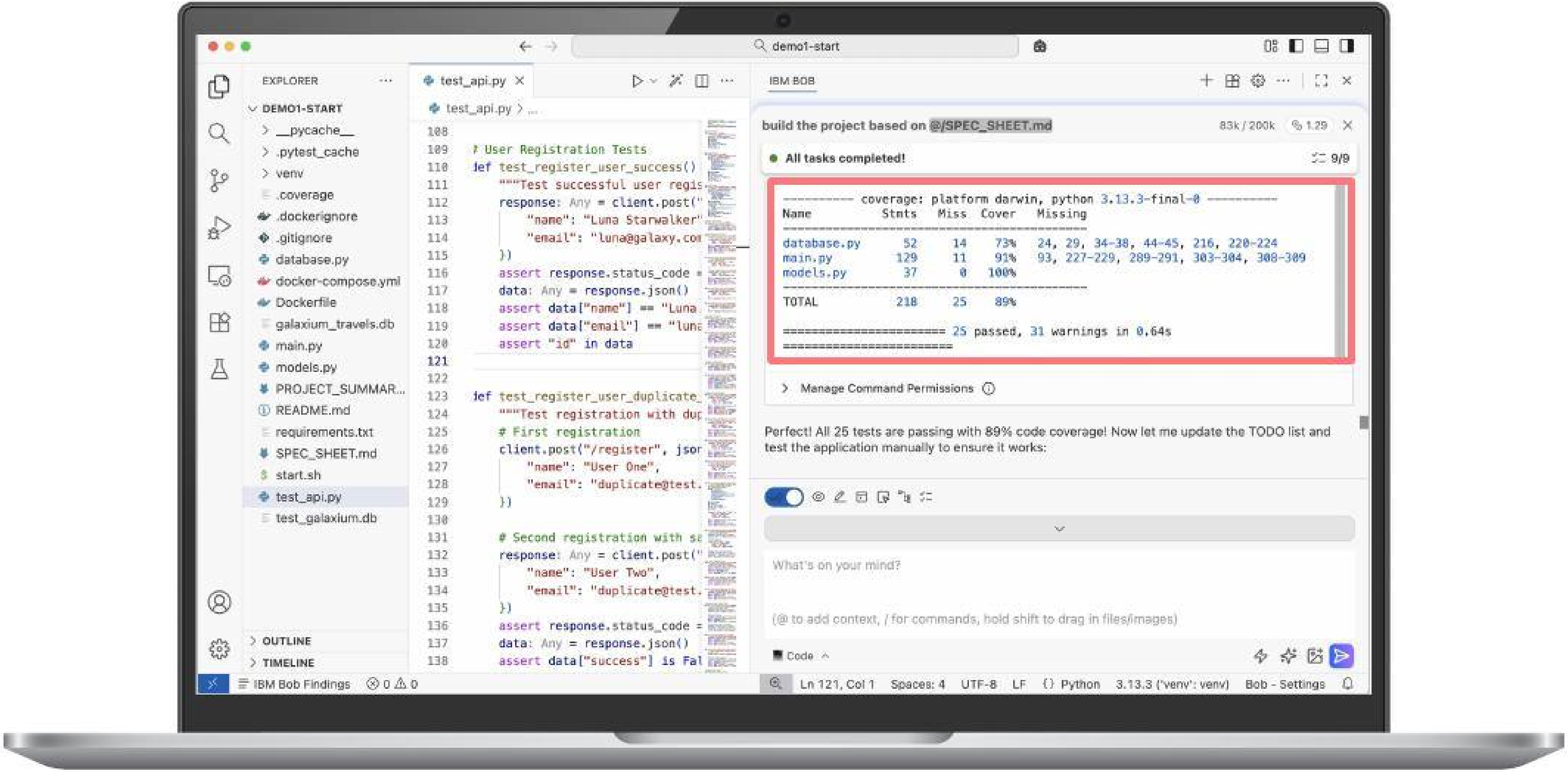


Less Steps

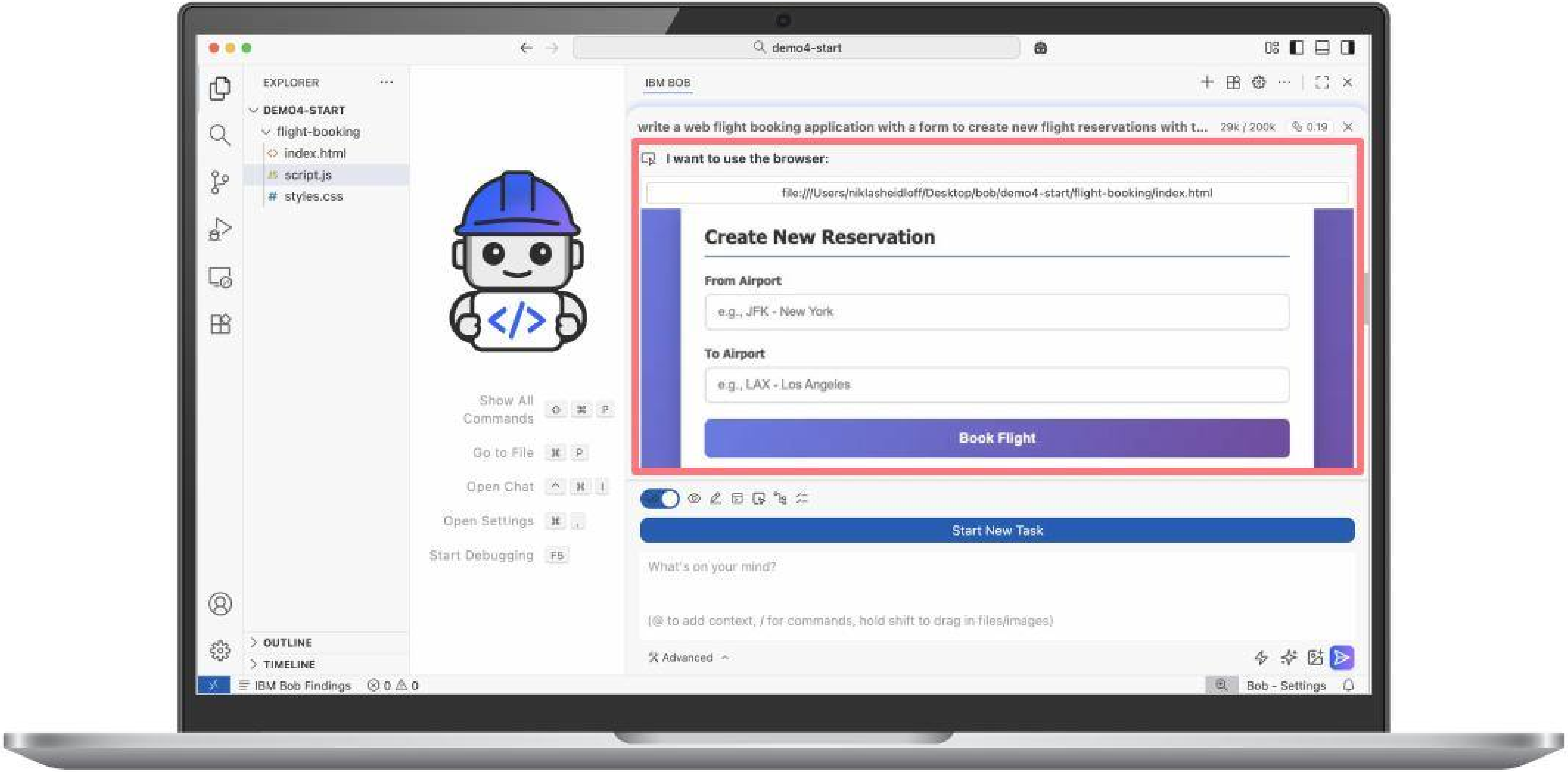


demo8-start
EXPLORER
DEMO8-START
.env.example
.gitignore
calculator_agen... 7
FILE_STRUCTURE.md
IMPLEMENTATION_...
PROJECT_SUMMAR...
QUICK_START.md
README.md
requirements.txt
TECHNICAL_SPEC...
test_agent.py
OUTLINE
TIMELINE
calculator_agent.py 7
"""
LangGraph Calculator Agent
A calculator agent that uses LangGraph to perform arithmetic ope
The agent uses an LLM to decide when to call calculator tools (a
"""
from langchain.tools import tool
from langchain_anthropic import ChatAnthropic
from langchain_core.messages import SystemMessage, HumanMessage,
from langgraph.graph import StateGraph, START, END
from typing_extensions import TypedDict, Annotated, Literal
import operator
import os
PROBLEMS 7
OUTPUT
Import "langchain.tools" could no... basedpyright(reportMissingImports) [Ln 8, Col 6]
Import "langchain_anthropic" cou... basedpyright(reportMissingImports) [Ln 9, Col 6]
Import "langchain_core.message... basedpyright(reportMissingImports) [Ln 10, Col 6]
Import "langgraph.graph" could ... basedpyright(reportMissingImports) [Ln 11, Col 6]
Import "dotenv" could not be res... basedpyright(reportMissingImports) [Ln 15, Col 6]
Type arguments for "Literal" ... basedpyright(reportInvalidTypeForm) [Ln 157, Col 67]
Import "typing_extension... basedpyright(reportMissingModuleSource) [Ln 12, Col 6]
IBM BOB
Hi, I'm IBM Bob
Ask me questions or let me code for you
Problems
Issues
Terminal
Paste URL to fetch contents
Add Folder
Add File
Git Commits
Git Branches
fix errors in @
Code
IBM Bob Findings
Refreshing virtual environments
Ln 1, Col 1 Spaces: 4 UTF-8 LF Python Bob - Settings

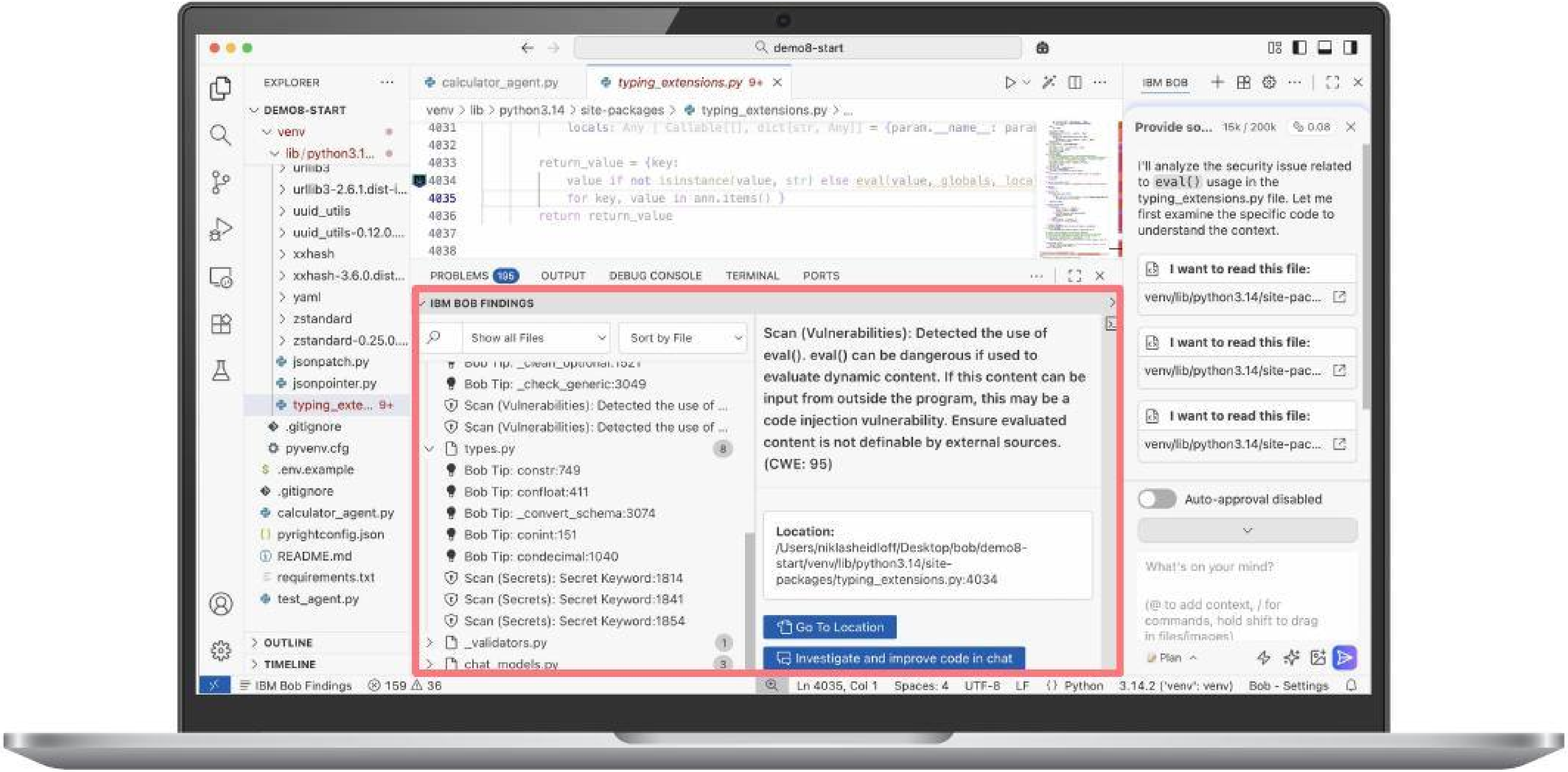
Integrated Testing



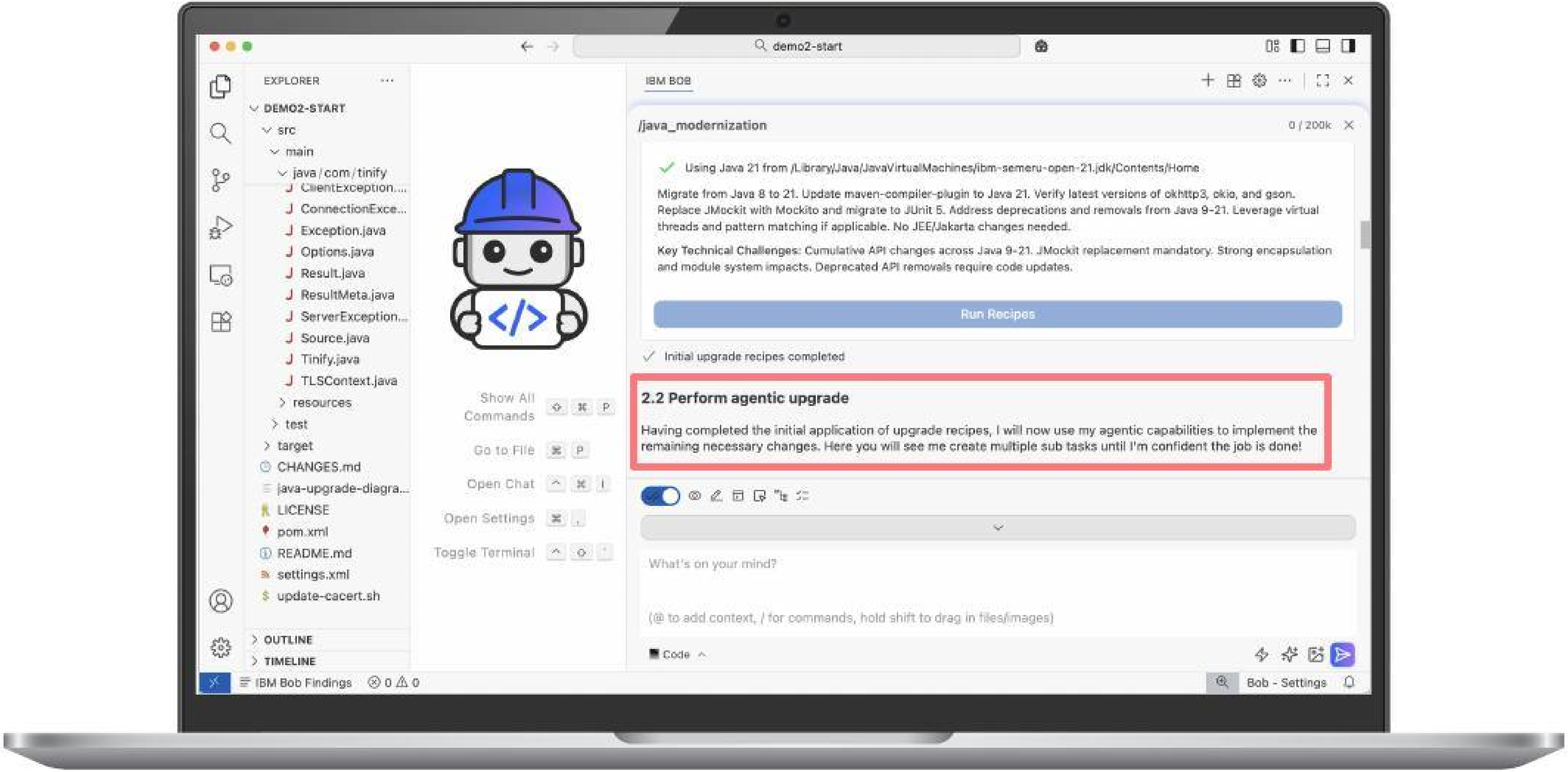
Integrated Testing



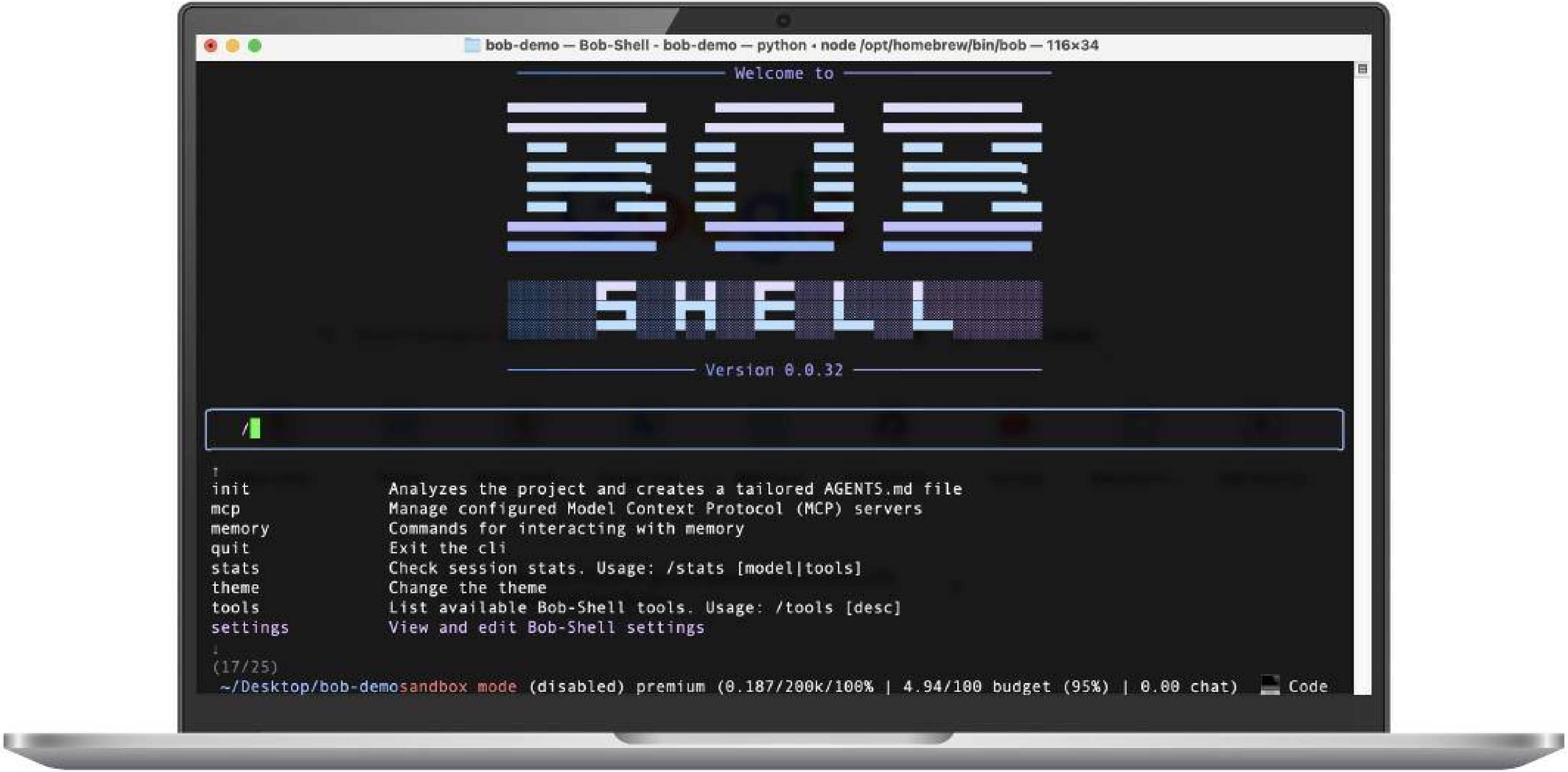
Shift Left Security



Combination of Rules and AI

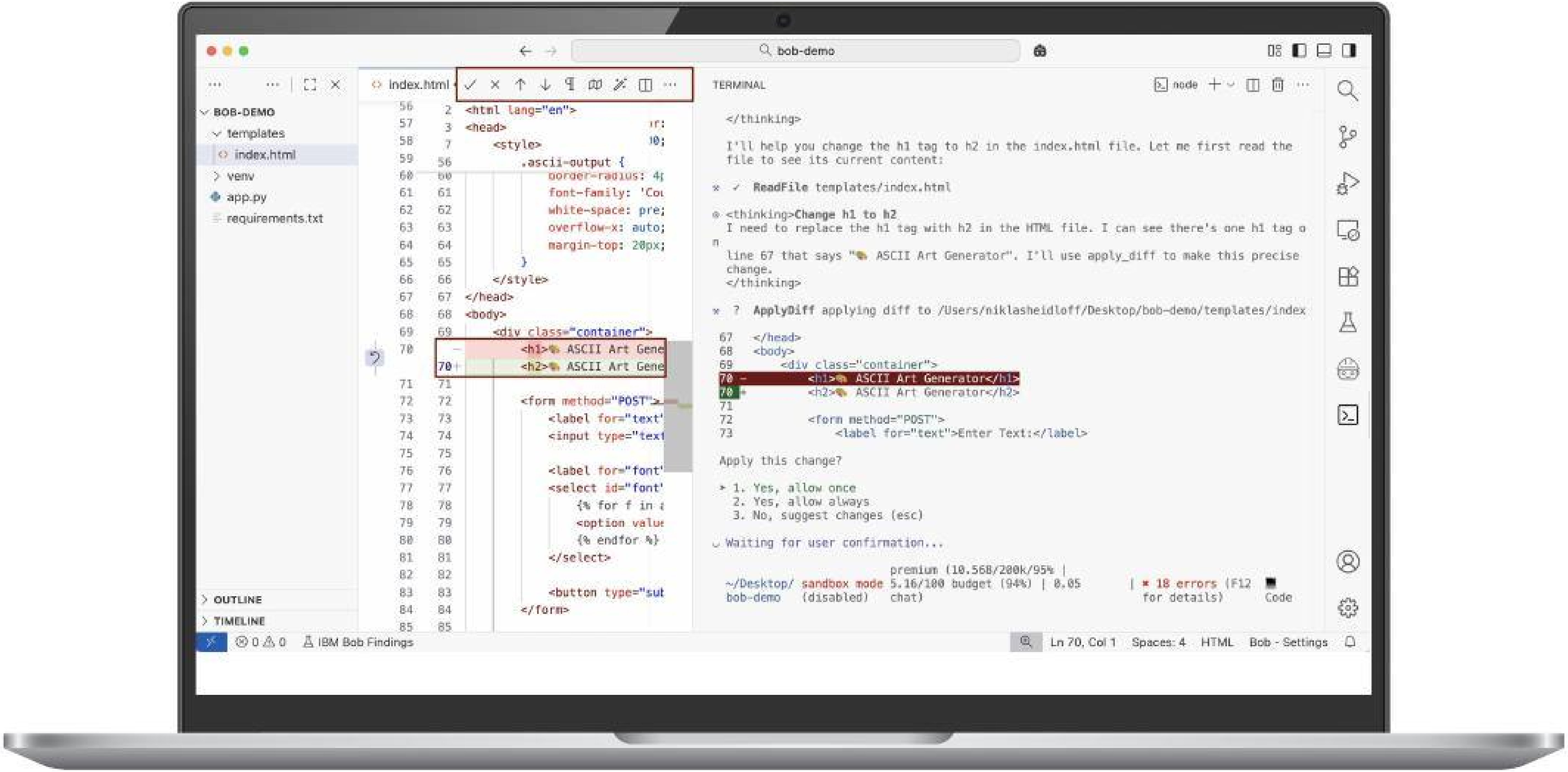


Bob Shell

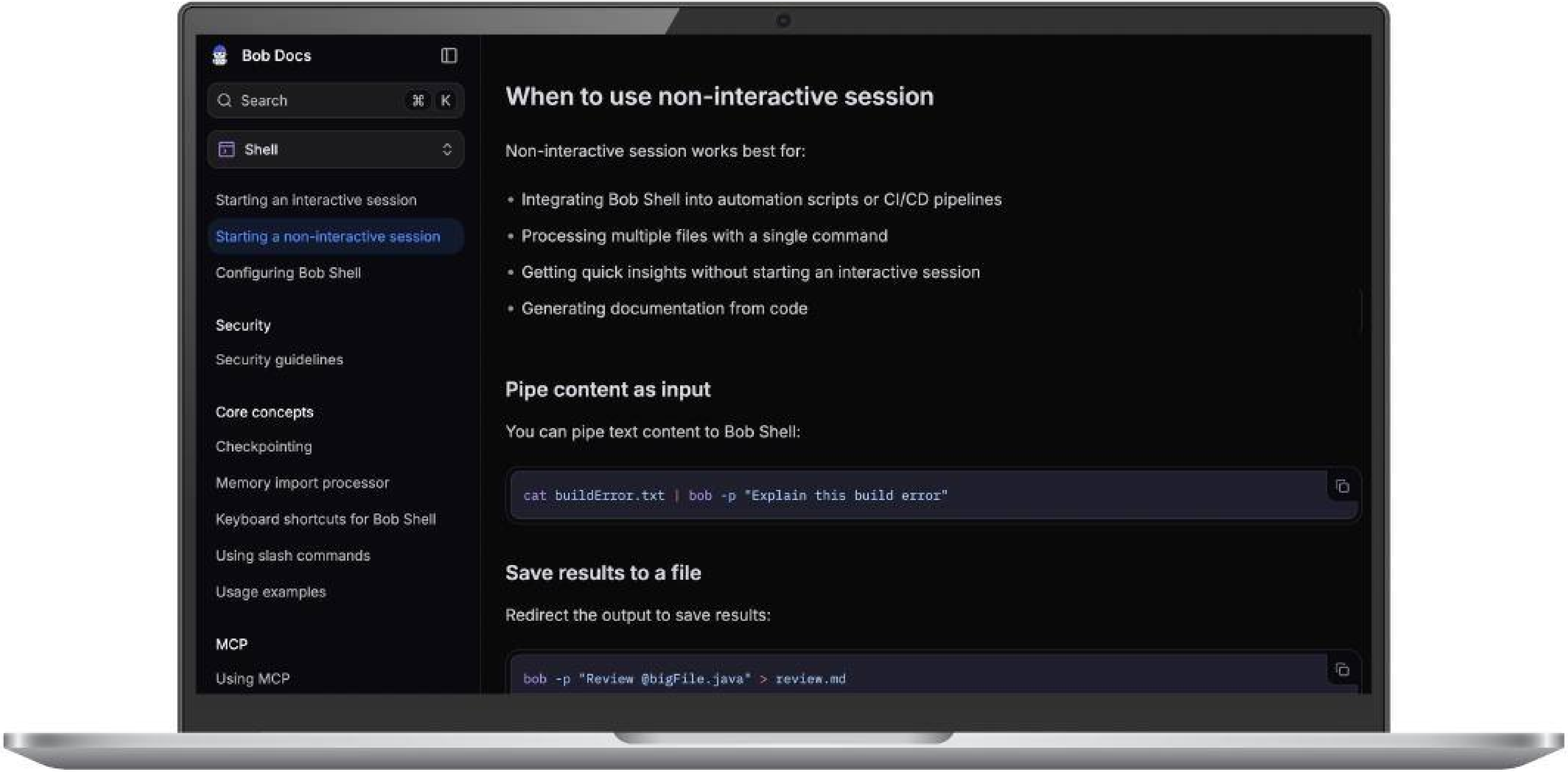


bob-demo — Bob-Shell - bob-demo — python • node /opt/homebrew/bin/bob — 116×34
Welcome to
BOB
SHELL
Version 0.0.32
/
init Analyzes the project and creates a tailored AGENTS.md file
mcp Manage configured Model Context Protocol (MCP) servers
memory Commands for interacting with memory
quit Exit the cli
stats Check session stats. Usage: /stats [model|tools]
theme Change the theme
tools List available Bob-Shell tools. Usage: /tools [desc]
settings View and edit Bob-Shell settings
(17/25)
~/Desktop/bob-demosandbox mode (disabled) premium (0.187/200k/100% | 4.94/100 budget (95%) | 0.00 chat)
Code

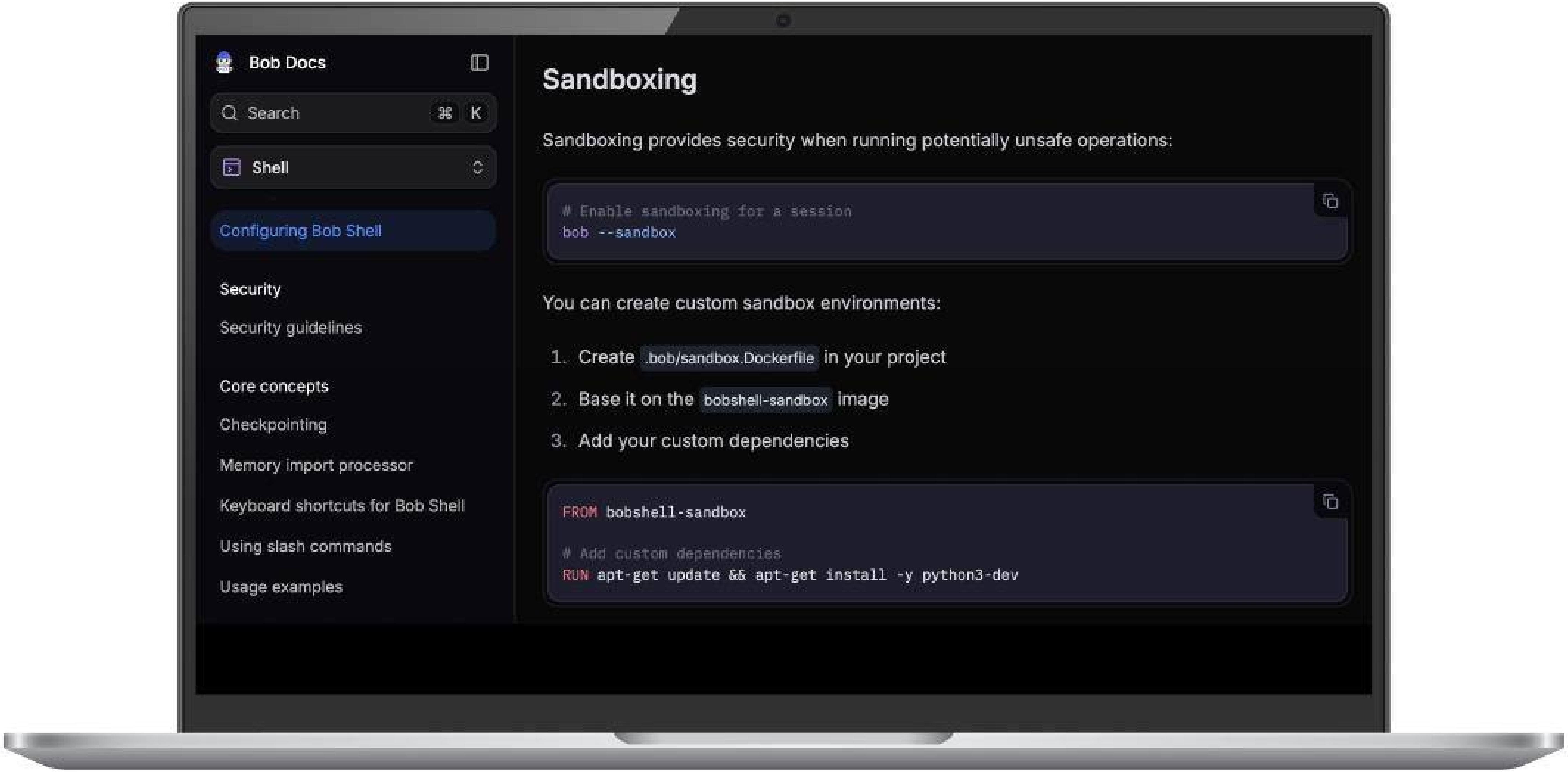
Bob Shell


bob-demo
index.html
BOB-DEMO
templates
index.html
venv
app.py
requirements.txt
OUTLINE
TIMELINE
TERMINAL
I'll help you change the h1 tag to h2 in the index.html file. Let me first read the file to see its current content:
✓ ReadFile templates/index.html
<thinking>Change h1 to h2
I need to replace the h1 tag with h2 in the HTML file. I can see there's one h1 tag on line 67 that says "ASCII Art Generator". I'll use apply_diff to make this precise change.
? ApplyDiff applying diff to /Users/niklasheidloff/Desktop/bob-demo/templates/index
Apply this change?
1. Yes, allow once
2. Yes, allow always
3. No, suggest changes (esc)
Waiting for user confirmation...
IBM Bob Findings
Ln 70, Col 1
Spaces: 4
HTML
Bob - Settings

Bob Shell


Bob Docs
Search
⌘ K
Shell
Starting an interactive session
Starting a non-interactive session
Configuring Bob Shell
Security
Security guidelines
Core concepts
Checkpointing
Memory import processor
Keyboard shortcuts for Bob Shell
Using slash commands
Usage examples
MCP
Using MCP
When to use non-interactive session
Non-interactive session works best for:
• Integrating Bob Shell into automation scripts or CI/CD pipelines
• Processing multiple files with a single command
• Getting quick insights without starting an interactive session
• Generating documentation from code
Pipe content as input
You can pipe text content to Bob Shell:
cat buildError.txt | bob -p "Explain this build error"
Save results to a file
Redirect the output to save results:
bob -p "Review @bigFile.java" > review.md

Bob Shell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Search
⌘ K
Shell
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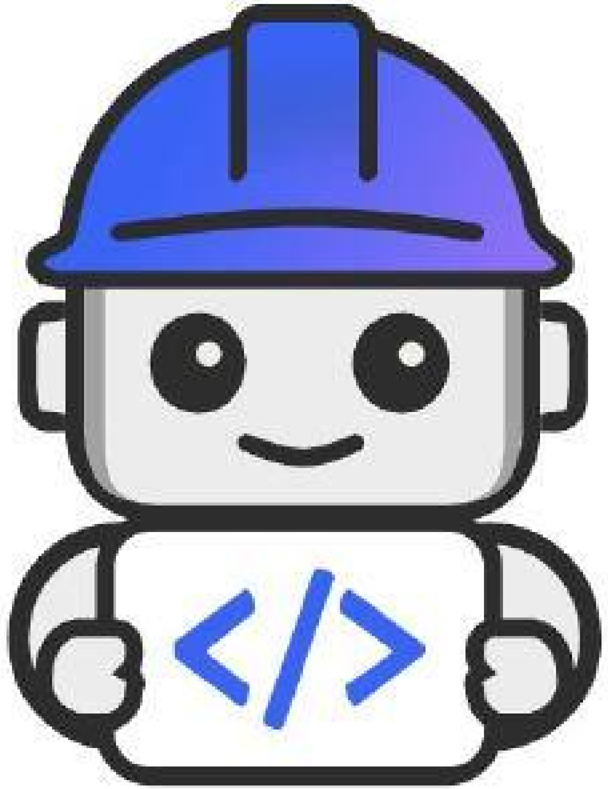
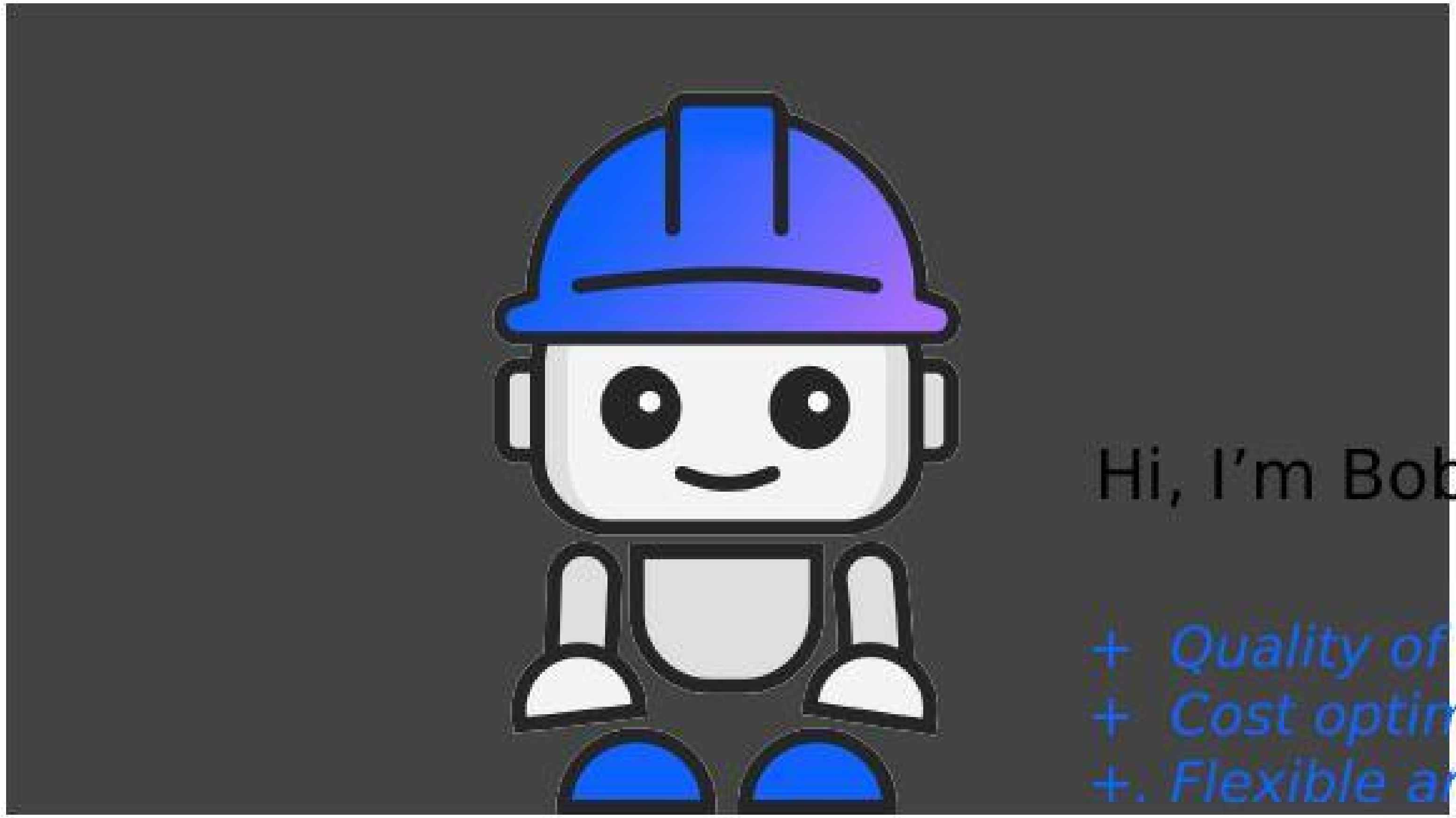
Security
Security guidelines
Core concepts
Checkpointing
Memory import processor
Keyboard shortcuts for Bob Shell
Using slash commands
Usage examples
Sandboxing
Sandboxing provides security when running potentially unsafe operations:
Enable sandboxing for a session
bob --sandbox
You can create custom sandbox environments:
1. Create .bob/sandbox.Dockerfile in your project
2. Base it on the bobshell-sandbox image
3. Add your custom dependencies
FROM bobshell-sandbox
Add custom dependencies
RUN apt-get update && apt-get install -y python3-dev

Key Concepts of IBM Bob

- Partnership instead of Automation
- Developer in Control
- Understanding before Coding
- Advancing established Development Processes
- Enterprise grade Code rather than Vibe Coding



Let’s try it out!



Hi, I’m Bob!

+ *Quality of output*
+ *Cost optimization and transparency*
+. *Flexible architecture*